

ORIGINAL RESEARCH ARTICLE

Activation of HIF2 in Cardiac Vasculature Leads to Arterial Remodeling, Dilation, Thrombosis, and Inflammation, Recapitulating Cardiac Involvement in Kawasaki Disease

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BACKGROUND: Global *Vhl* knockout results in vascular defects and early lethality, limiting our knowledge of VHL (von Hippel-Lindau)/HIF (hypoxia-inducible factor) signaling in coronary vessel formation and homeostasis. The hypoxia pathway has been implicated in cardiovascular diseases (CVDs) characterized by inflammation and vascular remodeling, such as atherosclerosis, but its involvement in Kawasaki disease (KD) remains unknown. Coronary artery dilation and vessel rupture are the most serious complications of KD. However, the molecular mechanisms underlying these cardiac events are not fully understood. We investigated the role of the VHL/HIF pathway in cardiovascular pathology and its relevance to KD.

METHODS: We generated a novel mouse model with genetic hyperactivation of the hypoxia pathway in progenitors contributing to coronary vessels and cardiac fibroblasts. We characterized the model using echocardiography, magnetic resonance imaging, histology, and molecular profiling. In parallel, we examined cardiac tissues from patients with KD with fatal coronary aneurysms for evidence of HIF signaling and inflammation using immunohistochemistry.

RESULTS: Mice with conditional deletion of *Vhl* in the *Wt1* (Wilms tumor 1) lineage developed normally but exhibited cardiomegaly, vascular abnormalities, progressive coronary artery dilation, pericardial hemorrhage, and systemic inflammation shortly after birth. Histologic analysis revealed coronary arteritis, elastin breaks, vascular remodeling, smooth muscle cell loss, perivascular fibrosis, and frequent intracoronary thrombus formation. In addition, vascular calcification, severe cardiac inflammation, and interstitial hemorrhages were observed, culminating in sudden death between 15 and 20 weeks of age, likely due to vessel rupture. Cardiac transcriptomic profiling identified dysregulated expression of genes involved in extracellular matrix organization, epithelial-mesenchymal transition, angiogenesis, inflammation, coagulation, and calcification, indicating compromised vascular stability and increased remodeling in *Vhl* conditional knockout mice. Simultaneous deletion of *Hif2a* rescued both the cardiovascular abnormalities and transcriptomic profile observed in *Vhl* conditional knockout mice, implicating Hif2 (hypoxia-inducible factor 2) as a key mediator. Human KD cardiac samples showed expression of HIF2 in coronary lesions and surrounding inflammatory infiltrates, confirming hypoxia pathway activation in severe KD.

CONCLUSIONS: Our findings establish HIF2 as a central driver of coronary inflammation, vascular remodeling, and thrombotic complications resembling those observed in severe KD. The *Vhl*/*Wt1* conditional knockout mouse model recapitulates key cardiovascular features of KD and offers a valuable platform for mechanistic studies and therapeutic exploration.

Key Words: aneurysm ■ HIP ■ hypoxia ■ Kawasaki disease ■ mucocutaneous lymph node syndrome ■ vasculitis ■ vascular remodeling ■ VHL

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Clinical Perspective

What Is New?

- Conditional deletion of *Vhl* in the Wt1 (Wilms tumor 1) lineage (*Vhl*/Wt1-Cre knockout) results in the sustained activation of Hif (hypoxia-inducible factor) signaling in coronary vessels and cardiac fibroblasts.
- Stabilization of Hif2 α in this model promotes coronary artery dilation, vascular remodeling, inflammation, thrombus formation, and calcification, recapitulating the major cardiovascular complications observed in severe forms of Kawasaki disease.
- Hif2 functions as a central transcriptional mediator of coronary vasculitis, and its genetic ablation rescues the cardiovascular pathology developed in *Vhl*/Wt1-Cre knockout mice.
- HIF2 α is expressed in coronary lesions and surrounding inflammatory infiltrates in cardiac tissue from patients with fatal Kawasaki disease, providing translational relevance to the murine findings.

What Are the Clinical Implications?

- HIF2 signaling may contribute to coronary inflammation, remodeling, and aneurysm formation in severe Kawasaki disease.
- Pharmacologic targeting of HIF2 may represent a novel therapeutic strategy to limit coronary inflammation and remodeling, thereby reducing the cardiovascular risk associated with human vasculitis, such as Kawasaki disease.

Cardiovascular diseases (CVDs) represent a leading cause of mortality and morbidity worldwide. Lifestyle-related risks, such as obesity, smoking, sedentary behavior, and hypertension, contribute to the burden of CVDs and are associated with sustained hypoxia in the heart. HIFs (Hypoxia-inducible factors) are central transcriptional regulators of the cellular response to low oxygen availability. Under normal conditions, HIF- α subunits are hydroxylated by PHDs (prolyl hydroxylases), which are oxygen sensors that target them for recognition by VHL (von Hippel–Lindau), which is part of an E3 ubiquitin ligase complex that facilitates HIF degradation through the proteasome. Hypoxia inhibits PHD activity, allowing HIF- α stabilization, dimerization with HIF- β , and subsequent activation of gene networks involved in angiogenesis, glycolysis, erythropoiesis, and extracellular matrix (ECM) remodeling, among others.¹

HIF activation is cardioprotective in ischemic injury, but chronic hypoxia is linked to pathologic cardiac remodeling. Sustained HIF activity, as seen in pressure overload, valvular dysfunction, or myocardial infarction, promotes maladaptive cardiac hypertrophy, contributing to the progression of heart failure.² In addition, hypoxia alters vascular homeostasis by stimulating angiogenesis,

Nonstandard Abbreviations and Acronyms

ACAN	aggrecan
ADAMT	a disintegrin-like and metalloprotease with thrombospondin type 1 motif
ADM	adrenomedullin
APLN	apelin
BDKRB2	bradykinin receptor B2
BMP	bone morphogenetic protein
CDH2	neural cadherin
CF	cardiac fibroblast
cKO	conditional knockout
CM	cardiomyocyte
CTSS	cathepsin S
CVD	cardiovascular disease
DKO	double knockout
EC	endothelial cell
ECM	extracellular matrix
FN1	fibronectin 1
HIF	hypoxia-inducible factor
HIF1	hypoxia-inducible factor 1
HIF2	hypoxia-inducible factor 2
IL-1β	interleukin-1 β
KD	Kawasaki disease
LOX	lysyl oxidase
LOXL2	lysyl oxidase-like 2
MCP-1	monocyte chemoattractant protein-1
MMP	matrix metalloproteinase
MPO	myeloperoxidase
MRP14	migration inhibitory factor-related protein 14
OPG	osteoprotegerin
PAI-1	plasminogen activator inhibitor 1
PHD	prolyl hydroxylase
POSTN	periostin
RANKL	receptor activator of nuclear factor κ B ligand
RNA-seq	RNA sequencing
SMA	smooth muscle actin
SMAD3	SMAD family member 3
SNAIL	snail family transcriptional repressor 1
TGF-β	transforming growth factor β
TIMP	tissue inhibitor of metalloprotease
TNFα	tumor necrosis factor- α
VHL	von Hippel–Lindau
VSMC	vascular smooth muscle cell
Wt1	Wilms tumor 1

increasing vascular permeability,³ and driving smooth muscle cell proliferation and migration, which can result in vascular remodeling and dysfunction.⁴

Experimental evidence implicates chronic HIF signaling in vascular diseases. HIF1 (hypoxia-inducible factor 1) has been associated with the development and dissection of age-related thoracic aortic aneurysms,⁵ and HIF2 (hypoxia-inducible factor 2) overexpression in vascular smooth muscle cells (VSMCs) derived from induced pluripotent stem cells increases ECM remodeling and cellular stiffness.⁶ In addition, induced pluripotent stem cell–derived endothelial cells (ECs) with HIF2 overactivation exhibit greater cell rigidity.⁶ Moreover, hypoxic regions within atherosclerotic plaques display upregulated HIF1 and HIF2, particularly in macrophage-rich thrombotic areas.⁷ Despite these insights, the role of HIFs in coronary vascular homeostasis in young individuals with vascular inflammation remains largely unknown.

VHL is essential for vascular development. Global *Vhl* knockout results in embryonic lethality due to placental and vascular defects.⁸ Sequence variations in the *Vhl* gene are associated with a greater incidence of developing highly vascularized tumors during adulthood, such as renal clear cell carcinomas, hemangioblastomas, and pheochromocytomas.⁹ Moreover, endothelial deletion of *Vhl* leads to aberrant vessel patterning and dilation during embryogenesis.¹⁰ Elimination of *Vhl* in the myocardium disrupts ventricular development and myocardial maturation, culminating in cardiac dysfunction and embryonic death.¹¹ These findings highlight the context-dependent roles of VHL in cardiovascular biology and underscore the need for lineage-specific models to assess its postnatal functions.

Several progenitor pools contribute to the structural and functional diversity of the adult heart.¹² Epicardial progenitors represent a mixed population with the potential to contribute to cardiac interstitial and adventitial fibroblasts, coronary smooth muscle cells, and, to a lesser extent, the coronary endothelium.^{13–16} Within the heterogeneity of epicardial precursors, Wt1 (Wilms tumor 1)–expressing progenitors contribute to the epicardium, interstitial cardiac fibroblasts (CFs), and adventitial and medial layers of the coronary vessels.¹⁴ Wt1 expression has also been detected in cardiac capillary ECs of both mice¹⁷ and humans,¹⁸ as well as in cardiac perivascular cells¹⁹ and a subset of embryonic cardiomyocytes (CMs).²⁰

Kawasaki disease (KD) is a rare pediatric systemic vasculitis that affects medium-sized vessels and is the leading cause of acquired CVD in children.²¹ KD predominantly affects children <5 years of age and can result in coronary artery aneurysms, dilation, thrombosis, and rupture, especially in untreated or treatment-delayed cases.²² Although a combination of genetic and environmental factors is believed to drive KD pathogenesis,^{23,24} its precise cause remains unclear. Animal models using *Lactobacillus casei*²⁵ or *Candida albicans* wall extracts^{26,27} have highlighted the importance of the innate immune response and proinflammatory cytokines, such as IL-1 β (interleukin-1 β) and TNF- α (tumor necrosis factor- α), in the development of vascular lesions.^{28,29} However, the

roles of hypoxia and HIF signaling in KD pathogenesis have not yet been investigated.

In this study, we developed a novel mouse model with *Vhl* deletion in the Wt1 lineage to investigate the effects of sustained HIF activation on the coronary vasculature and cardiac function. These mice exhibited sparse stabilization of Hif1 and extensive expression of Hif2, accompanied by coronary artery dilation, inflammation, remodeling, thrombosis, calcification, and ultimately sudden cardiac death, recapitulating the key features of severe KD. Simultaneous deletion of *Vhl* and *Hif2 α* in the Wt1 lineage rescued all vascular and cardiac abnormalities observed in the *Vhl*/Wt1 conditional knockout (cKO) model (*Vhl* cKO). Moreover, analysis of post-mortem cardiac tissue from patients with KD revealed HIF2 activation in coronary lesions, providing evidence of hypoxia signaling involvement in human disease. Our findings collectively reveal a previously unrecognized role for HIF2 in coronary vasculitis and establish the novel *Vhl*/Wt1 cKO model as a valuable tool for elucidating the molecular mechanisms underlying KD-associated cardiovascular complications.

METHODS

Expanded [Methods](#) can be found in the Supplemental Material.

Data Availability

The data supporting this study are available from the corresponding author upon reasonable request. RNA sequencing (RNA-seq) data are available in the Gene Expression Omnibus under accession number GSE312198.

Animal Care and Housing and Human Material

Vhl flox/flox³⁰ and *Hif2 α* flox/flox³¹ mice were obtained from Jackson Laboratories (strains 004081 and 008407, respectively), maintained on a C57BL/6J background (including crosses with the C57BL/6JOLA^{Hsd} substrains), and crossed with mice heterozygous for the Wt1-Cre recombinase transgene³² (C57BL/6J background). Mice were housed under specific pathogen-free conditions at the CNIC Animal Facility. All procedures involving animals conformed to EU Directive 2010/63EU and Recommendation 2007/526/EC, as implemented in Spanish law under Real Decreto 53/2013. All mouse experiments were approved by the CNIC Animal Experimentation Ethics Committee and the competent national authorities (PROEX 267/19 and PROEX 104.8/25).

Human cardiac tissue portions were obtained from autopsy samples of patients with KD collected between 1970 and 1991 and stored at Toho University, Faculty of Medicine, and 3 medical centers in Japan. Five samples were anonymized using a correspondence table to protect the identification of individuals. Information accompanying the samples includes age (3, 7, 8, 9, and 36 months old), sex (2 female and 3 male), and the interval between the onset of KD and death (10, 15, 18, 23, and 30 days). Immunofluorescence and immunohistochemistry analyses of HIF factors in these tissues, together with evaluation of the nature of the inflammatory infiltrate, were approved by the Human Ethics Committee of Toho University Ohashi

Medical Center, Tokyo, Japan, in April 2020 (institutional review board approval H20006).

Statistical Analysis and Data Representation

Sample size estimation was not set a priori but based on our previous experience with comparable mouse models to achieve $\approx 90\%$ statistical power at a significance level of 0.05, while adhering to the principles of the 3Rs (replacement, reduction, and refinement) to minimize animal use. Sample sizes ranged from 3 to 20 animals per group depending on the experiment and availability of genetically modified animals. No animals were excluded from the study unless removal was required due to ethical end point criteria.

Statistical analyses were performed using GraphPad Prism version 10 software. All data are presented as mean $\pm$ SEM unless otherwise indicated. Each data point represents an independent biologic replicate (individual animal). Data were obtained from ≥ 3 independent experiments using animals from each experimental group derived from ≥ 3 independent litters.

For data sets with ≥ 10 animals per group, the normality of data distribution was assessed using the Shapiro-Wilk test. For normally distributed data, comparisons between 2 groups were made using an unpaired 2-tailed Student *t* test, and multiple comparisons were analyzed using 1-way ANOVA followed by the Tukey post hoc test. For non-normally distributed data, non-parametric tests were applied (the Mann-Whitney *U* test for 2 groups or the Kruskal-Wallis test followed by the Dunn post hoc test for multiple comparisons).

For experiments with < 10 animals per group, formal tests of normality have limited statistical power and cannot reliably detect deviations from normality. For these data sets, we applied non-parametric tests (the Mann-Whitney *U* test for 2 groups or the Kruskal-Wallis test followed by the Dunn post hoc test for multiple comparisons). In these cases, *P* values reaching the lower limit given the sample size are indicated in the corresponding figure legends (eg, 0.100* represents the lowest attainable *P* value for $n_1=3$ and $n_2=3$ or 0.0571 for $n_1=3$ and $n_2=4$). Furthermore, visual inspection of the data showed highly consistent “all-or-none” effects with no outliers in these cases, reflecting the biologic significance of the result despite the sample size limitation.

Survival analysis was performed using the log-rank (Mantel-Cox) test, and differences in categorical variables were assessed using the χ^2 test. For statistical analysis of RNA-seq data sets, multiple testing correction was applied using the Benjamini-Hochberg procedure to control the false discovery rate. Adjusted *P* values (*q* values) < 0.05 were considered statistically significant. Samples were analyzed in a blinded manner by different researchers, and experimental groups were randomly assigned to each assay. Representative images were selected to illustrate the typical phenotypes observed for each experimental group.

RESULTS

Lack of *Vhl* in Wt1 Coronary Progenitors Does Not Cause Developmental Abnormalities but Leads to Vascular Defects, Cardiomegaly, and Reduced Postnatal Survival

To investigate the role of VHL in the establishment of coronary vessels during cardiogenesis, we generated a

novel conditional mouse model by crossing *Vhl*-floxed mice³⁰ with Wt1-Cre mice,³² hereafter referred to as *Vhl* cKO mice. Histologic analysis of *Vhl* cKO hearts at embryonic day (E) 18.5 did not reveal alterations in coronary vessel patterning or diameter, or in ventricular chamber shape, compared with littermate controls (Figure 1A).

Analysis at weaning indicated that *Vhl* cKO mice survived embryogenesis and were recovered at the expected Mendelian frequencies (Figure 1B). No significant differences were observed in the percentage of recovery of *Vhl* cKO male mice (22.04% $\pm$ 2.91 [$n=24$ litters]) versus *Vhl* cKO female mice (21.33% $\pm$ 2.63 [$n=24$ litters]). Efficient deletion of *Vhl* exon 1 was confirmed by polymerase chain reaction (Figure S1A), and was associated with reduced *Vhl* mRNA levels and increased expression of the Hif target genes *Glut1*, *Phd3*, and *Pai1* in *Vhl* cKO hearts compared with controls (Figure S1B).

Heterozygous deletion of *Vhl* in the Wt1 lineage did not result in structural cardiac defects or decreased viability (data not shown), but *Vhl* homozygous cKO mice exhibited significantly reduced survival among both males and females, with a median lifespan of 11.2 weeks (Figure 1C). Whole-mount organ examination revealed cardiomegaly, right atrial dilation, and prominent vascular lesions in *Vhl*-deficient mice (Figure 1D). The heart weight–body weight ratio was consistently significantly increased in *Vhl* cKO mice (Figure 1E). Nonsignificant differences were observed in heart weight–body weight ratio between *Vhl* cKO male mice (0.9 $\pm$ 0.07 [$n=9$]) and *Vhl* cKO female mice (0.8 $\pm$ 0.07 [$n=7$]).

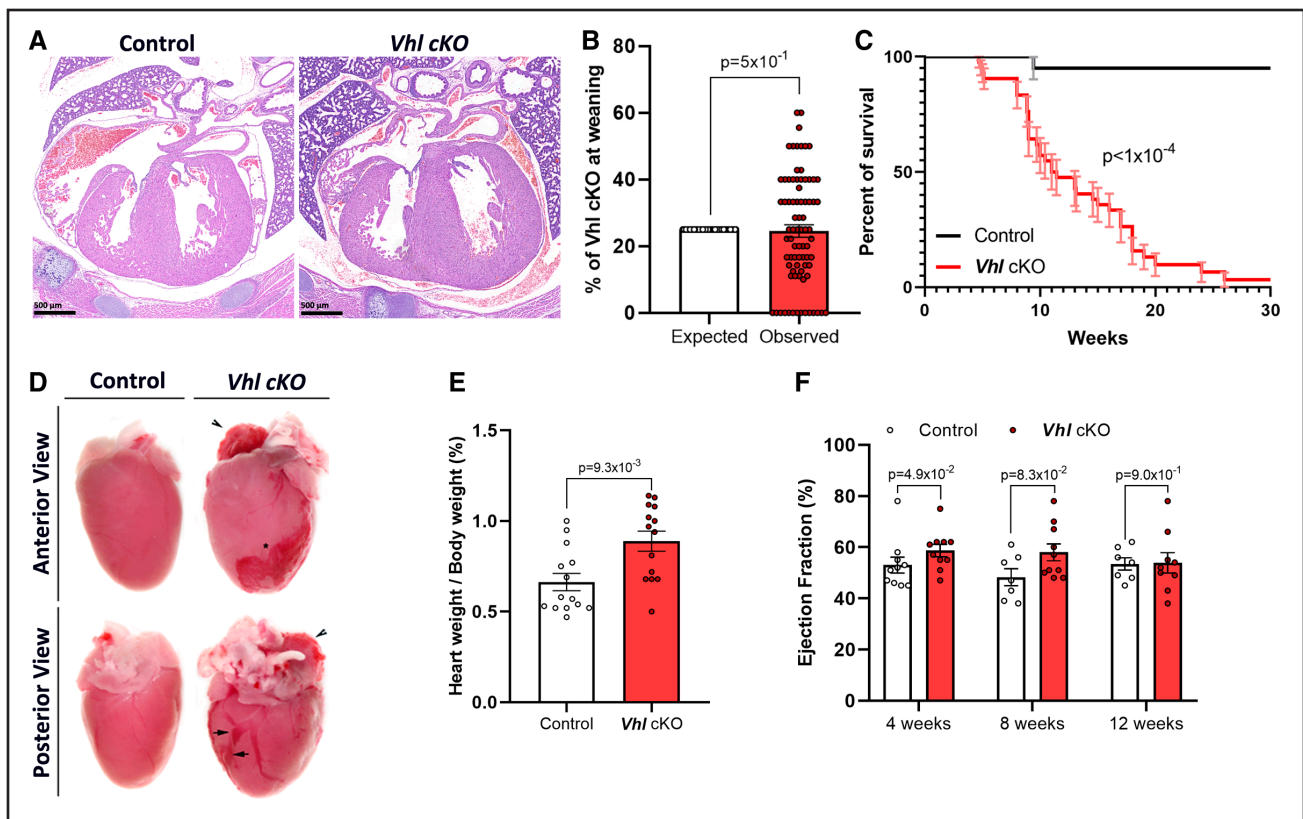
Despite these structural abnormalities, systolic function remained preserved in *Vhl* cKO mice, as evidenced by the normal ejection fraction (Figure 1F) and fractional shortening (Figure S2A). In line with these results, the early to late ventricular filling velocity ratio, a marker of diastolic dysfunction, showed no significant elevation in *Vhl* cKO mice compared with controls, despite a slight increase by 12 weeks of age (Figure S2B).

Increased ventricular dilation at 12 weeks, as measured by left ventricular volume (Figure S2C) and left ventricular inner diameter (data not shown), may underline the cardiomegaly phenotype observed in *Vhl* cKO mice relative to controls.

These findings indicate that epicardial *Vhl* is dispensable for normal coronary vessel development during embryogenesis but *Vhl* deletion results in acquired cardiac pathologies early in postnatal life both in male and female mice.

Activation of HIF Signaling Triggers Progressive Coronary Dilation and ECM Remodeling

In addition to the prominent vascular lesions observed in whole-mount *Vhl* cKO hearts (Figure 1D), echocardiographic analysis revealed significant pericardial



hemorrhage in *Vhl* cKO mice compared with controls (data not shown), suggestive of coronary vascular fragility. This observation was further confirmed by magnetic resonance imaging, which also demonstrated enlarged cardiac size and abnormally dilated coronary arteries in 8-week-old *Vhl* cKO mice (Figure 2A) with respect to controls. Complementary echocardiography revealed coronary artery dilation in *Vhl* cKO mice compared with controls, without concurrent aortic abnormalities (Figure 2B). Longitudinal echocardiographic analysis from postnatal day (P) 9 to 12 weeks of age demonstrated a progressive increase in coronary diameter in *Vhl* cKO mice in comparison with controls (Figure 2C). Nonsignificant differences were observed in the coronary diameter of *Vhl* cKO male mice versus *Vhl* cKO female mice at 4 weeks of age (0.22 ± 0.03 [$n=5$] versus 0.36 ± 0.17 [$n=4$]) or 8 weeks of age (0.34 ± 0.05 [$n=7$] versus 0.32 ± 0.08 [$n=5$]).

Histologic analysis using hematoxylin & eosin staining confirmed that *Vhl* cKO pups at P15 already exhibited coronary artery remodeling relative to controls, which progressed to marked dilation by 8 weeks of age, particularly in the proximal right coronary artery (Figure 2D). In contrast, the aortic medial layer remained structurally intact in *Vhl* cKO mice (Figure S3), consistent with the fact that VSMCs in the aorta do not derive from the *Wt1* lineage.

Given that VHL is a negative regulator of HIFs, and that the epicardial *Wt1* lineage gives rise to coronary ECs, VSMCs, and perivascular fibroblasts (Figure S4),¹⁹ we assessed HIF expression in the coronary vasculature. Immunofluorescence analysis revealed scattered Hif1 α expression in ECs, VSMCs, and the adventitial area of the dilated coronary arteries in *Vhl* cKO mice compared with controls (Figure S5A). Moreover, Hif2 α expression was markedly increased in coronary VSMCs, the endothelium,

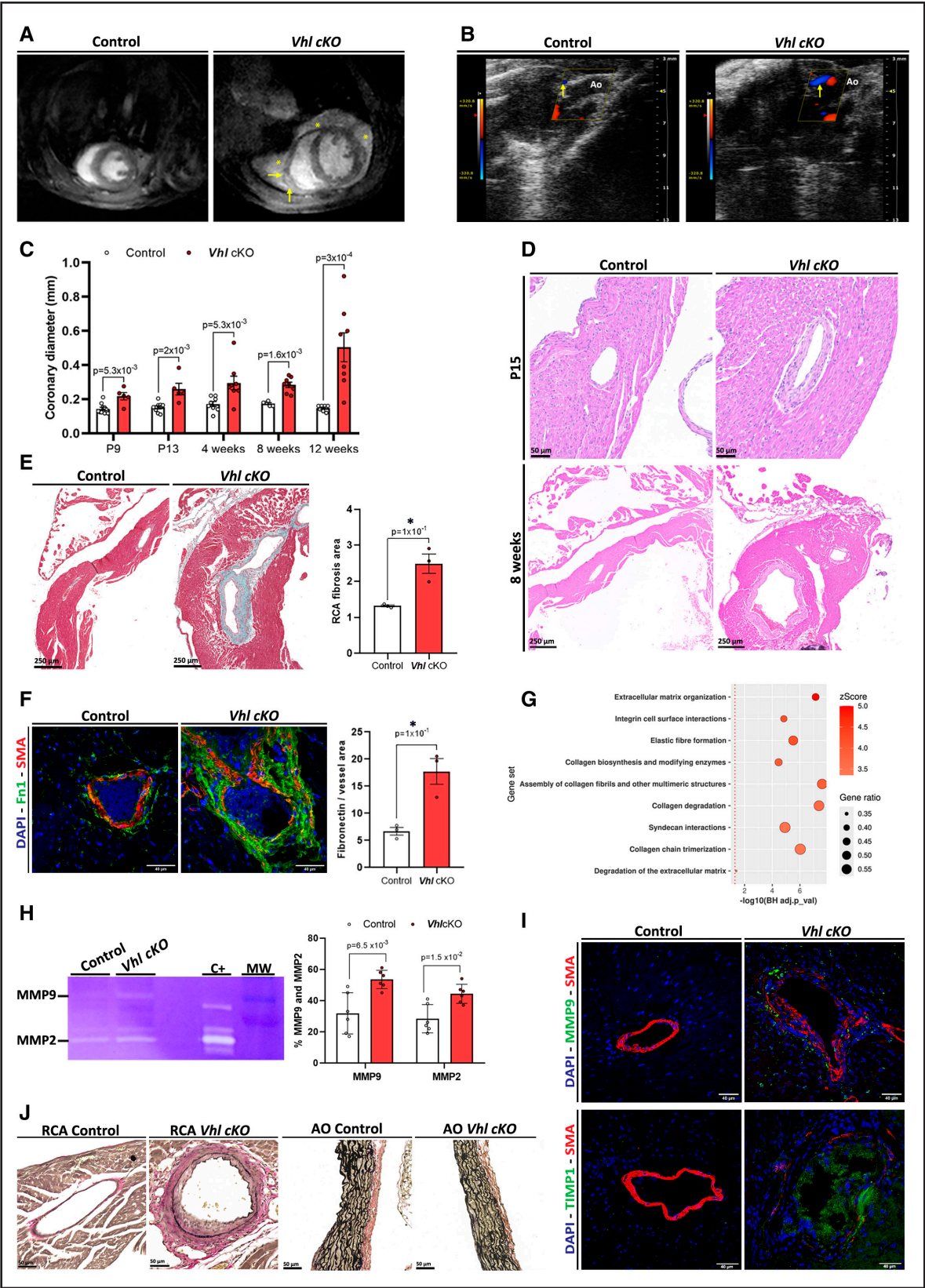


Figure 2. Activation of Hif signaling triggers progressive coronary dilation and enhances extracellular matrix remodeling. **A**, Representative magnetic resonance imaging showing an enlarged heart in the *Vhl* conditional knockout (cKO) mice with dilated coronaries (yellow arrows) and pericardial hemorrhage (yellow asterisks) relative to the control. **B**, Illustrative echocardiographic imaging revealed prominent enlargement of the coronary vessels (yellow arrows) in the *Vhl* cKO mice without alterations in the aorta (Ao) compared with littermate controls. **C**, Longitudinal quantification of the coronary diameter from postnatal day 9 (P9) to 12 weeks of age measured by echocardiographic (*Continued*)

Figure 2 Continued. analysis showing progressive enlargement in the *Vhl* cKO mice (red bars) compared with controls (white bars). Control: P9 (n=9), P13 (n=9), 4 weeks (n=8), 8 weeks (n=5), 12 weeks (n=7); *Vhl* cKO: P9 (n=5), P13 (n=5), 4 weeks (n=8), 8 weeks (n=8), 12 weeks (n=8). **D**, Histologic analysis by hematoxylin & eosin staining of ventricular cardiac sections from control and *Vhl* cKO mice on P15 (top panels) or 8 weeks of age (bottom panels) in the region of the right coronary artery (RCA). Scale bars, 50 μ m (P15) and 250 μ m (8 weeks). **E**, Representative Masson trichrome staining of heart sections from 8-week-old control and *Vhl* cKO mice showing collagen deposition (blue) in the coronary vessels from the right ventricle. Scale bars, 250 μ m. Quantification of the fibrotic area in the RCA of control (white bars) and *Vhl* cKO (red bar) mice is shown (n=3). **F**, Illustrative immunofluorescence analysis of Fn1 (fibronectin 1; green), SMA (smooth muscle actin; red), and DAPI (blue) in cardiac sections from 8-week-old control and *Vhl* cKO mice at the level of a coronary artery. Scale bars, 40 μ m. Quantification of Fn1 signal per coronary area in control (white bars) and *Vhl* cKO mice (red bars) is shown (n=3). **G**, Dot plot of selected biofunctions detected as enriched by Ingenuity Pathway Analysis in the set of differentially expressed genes associated with the contrast *Vhl* cKO versus control. The dot plot presents the results for terms related to extracellular matrix. Enrichment significance is represented on the x axis as $-\log_{10}$ (Benjamini-Hochberg P_{adjusted}). The dot color represents the z score (positive values indicate higher activity in *Vhl* cKO mice), and the dot size represents the gene ratio (fraction of function-associated genes present in the focus gene set). **H**, Representative zymography assay from native cardiac ventricular lysates of 8-week-old control and *Vhl* cKO mice showing activity levels of MMP9 (matrix metalloproteinase 9) and the active form of MMP2 (matrix metalloproteinase 2). C⁺ represents a positive control from an aortic lysate of a Marfan syndrome mouse model. Zymography quantification representing the percentage of MMP9 and MMP2 activities in control (white bars) and *Vhl* cKO (red bars) mice is shown (n=6). **I**, Representative immunofluorescence images of MMP9 (green, upper panels), TIMP1 (tissue inhibitor of metalloprotease; green, bottom panels), SMA (red), and DAPI (blue) in cardiac ventricular sections from 8-week-old control and *Vhl* cKO mice at the level of a coronary artery. Scale bars, 40 μ m. **J**, Representative Verhoeff–Van Gieson staining showing elastin (black lines) and collagen (pink) in cardiac ventricular sections from control and *Vhl* cKO mice by 8 weeks of age at the level of a coronary artery (left panels) showing discontinuity in the elastic lamina of the remodeled artery in the *Vhl* cKO mice. The elastin pattern at the level of the aorta (AO, right panels) in *Vhl* cKO mice is comparable to controls. Scale bars, 50 μ m. Bars represent mean $\pm$ SEM. Statistical significance was evaluated using nonparametric Mann-Whitney *U* tests. The asterisks in **E** and **F** indicate that 0.100 is the lowest attainable *P* value given the sample size. At least 3 independent animals per genotype from separate litters were used for each analysis. Hif indicates hypoxia-inducible factor; MW, molecular weight markers; and *Vhl*, von Hippel–Lindau.

and, to a lesser extent, the adventitia of altered vessels relative to controls (Figure S5B).

To assess the structural abnormalities associated with coronary artery dilation, we examined ECM remodeling in *Vhl* cKO mice. Masson trichrome staining of cardiac sections revealed markedly increased perivascular collagen deposition in the dilated ventricular coronary arteries of *Vhl* cKO mice compared with controls, suggestive of fibrosis and changes in matrix architecture (Figure 2E), with no obvious qualitative differences in the aorta (data not shown). Immunofluorescence staining for Fn1 (fibronectin 1), a key ECM component, showed increased expression in the dilated ventricular coronaries of *Vhl* cKO hearts compared with the control group (Figure 2F).

To gain mechanistic insight, we performed bulk RNA-seq of ventricular cardiac tissue from 8-week-old *Vhl* cKO and control mice. A total of 12 760 protein-coding genes were detected, 2085 of which were differentially expressed genes between *Vhl* cKO and control mice (Benjamini-Hochberg $P_{\text{adjusted}} < 0.05$; Table S1 and Figure S6). Ingenuity Pathway Analysis confirmed the expected altered expression of genes related to the classic cellular response to hypoxia, including the downregulation of oxidative phosphorylation, mitochondrial complex assembly, and protein translation (Figure S7).

Functional analysis of upregulated genes in *Vhl* cKO hearts relative to controls revealed significant enrichment (Benjamini-Hochberg $P_{\text{adjusted}} < 0.05$) of pathways involved in collagen biosynthesis, degradation, and cross-linking; elastic fiber formation; ECM organization; and syndecan/integrin interactions (Figure 2G and Table S2). We identified increased expression of several collagens, including *Col12a1* or *Col18a1*, which are key subendothelial components that support VSMC migration and

proliferation and vessel wall integrity (Tables S1 and S2). The expression of lysyl oxidases and lysyl oxidase-like enzymes, involved in collagen and elastin crosslinking, was also elevated, along with Ctss (cathepsin S), a lysosomal cysteine proteinase capable of degrading elastin, fibronectin, and collagens (Tables S1 and S2).

Moreover, several members of the MMP (matrix metalloproteinase) family, TIMPs (tissue inhibitor of metalloproteinases), and ADAMTs (A disintegrin-like and metalloprotease with thrombospondin type 1 motif), which are enzymes involved in ECM remodeling and modulation of inflammatory mediators, were significantly dysregulated in *Vhl* cKO mice compared with controls (Tables S1 and S2). To functionally validate ECM degradation, we performed zymography, which showed increased MMP9 protein levels and enhanced activity of the cleaved, active form of MMP2 in *Vhl* cKO hearts in relation to controls (Figure 2H). Both MMP9 and MMP2 exhibit proteolytic activity toward collagens, fibronectin, and elastin, and have been previously described to be modulated by hypoxia.³³ Immunofluorescence further confirmed elevated MMP9 expression between the adventitial and medial layers of dilated coronary vessels in *Vhl* cKO mice versus controls (Figure 2I, upper panels). Moreover, we confirmed increased gene and protein expression of TIMP1 in altered coronaries of *Vhl* cKO mice relative to controls (Table S1 and Figure 2I, lower panels).

We also assessed elastic fiber organization using Verhoeff–Van Gieson staining. In line with the transcriptomic findings, *Vhl* cKO coronary arteries displayed fragmented and discontinuous elastic laminae, particularly in regions of medial thickening and vascular hyperplasia, in contrast with the conserved structures observed in controls

(Figure 2J, left panels). These alterations were not observed in the aortas of *Vhl* cKO mice, which exhibited normal elastin architecture as control aortas (Figure 2J, right panels).

These results collectively demonstrate that loss of *Vhl* in *Wt1* epicardial progenitors leads to progressive coronary artery dilation driven by HIF stabilization and pathologic remodeling of the ECM.

Elimination of *Vhl* in the *Wt1* Lineage Induces Cardiac Damage, Capillary Hemorrhages, and Fibrosis

Consistent with previous reports,²⁰ lineage tracing analysis of *Wt1*-derived cells in adult mice revealed a positive signal in scattered CMs of the interventricular septum, papillary muscles, and both ventricular walls (Figure S8A–S8D), in addition to the contribution to the microvasculature in these regions.¹⁹ Analysis of these structures by Masson trichrome staining in 4-week-old *Vhl* cKO mice in comparison with controls revealed early signs of cardiomyocyte damage and microvascular abnormalities, including remodeling of the surrounding capillaries (Figure S8E–S8H).

By 8 weeks of age, histologic characterization of *Vhl* cKO hearts showed evident tissue remodeling relative to controls, with dilated hemorrhagic capillaries and increased interstitial fibrosis, particularly in the papillary muscles and interventricular septum (Figure 3A). In addition, increased collagen deposition was detected along the epicardium of both ventricles and atria in *Vhl* cKO mice relative to controls (Figure S9A). Quantification of Masson trichrome staining revealed an increased percentage of fibrotic areas in *Vhl*-deficient hearts compared with controls (Figure 3B). This accumulation of collagen is consistent with the development of patches of interstitial fibrosis, which may contribute to the trend toward diastolic dysfunction observed in 12-week-old *Vhl* cKO mice (Figure S2B). Furthermore, this fibrotic pattern correlates with the altered expression of genes involved in ECM organization and remodeling, as well as collagen biosynthesis, assembly, and degradation (Table S2 and Figure 2G).

Longitudinal electrocardiographic analysis revealed a trend toward prolonged PR interval (Figure S9B) and QRS duration (Figure S9C) in *Vhl* cKO mice compared with controls by 12 weeks of age. These results suggest that despite displaying increased fibrosis, *Vhl* cKO mice do not develop electrophysiologic abnormalities.

Functional analysis of the cardiac transcriptome revealed significant enrichment of processes related to fibrosis, abnormal connective tissue development, and epithelial–mesenchymal transition in *Vhl* cKO mice compared with controls (Figure 3C). This included upregulation of *Cdh2* (neural cadherin [*Cdh2*]), *Snai1* (snail family transcriptional repressor 1 [*Snai1*]), *Postn* (peri-

ostin [*Postn*]), and *Lox* (lysyl oxidase [*Lox*]) or *Loxl2* (lysyl oxidase-like 2 [*Loxl2*]) genes, all of which have been previously implicated in cell matrix adhesion, epithelial–mesenchymal transition, ECM remodeling, and aneurysm formation (Table S2). Furthermore, gene set enrichment analysis identified enrichment of the BMPs (bone morphogenetic proteins) and TGF- β (transforming growth factor β) signaling pathways, which are well-known mediators of fibrosis and vascular remodeling (Figure 3D), with *Wnt3a* and *SMAD3* (SMAD family member 3) appearing among the top predicted upstream regulators (Table S2). We confirmed elevated protein levels of several induced genes related to fibrosis and ECM remodeling, such as *Postn*, *Acan* (aggrecan), and *Bmp2*, in the altered cardiac tissue of *Vhl* cKO mice relative to that in control hearts (Figure S10).

Next, we assessed Hif activation at the papillary muscles contributed by *Wt1* (Figure S8) by immunofluorescence in 8-week-old hearts. Both Hif1 α (Figure S11A) and Hif2 α (Figure S11B) levels were significantly elevated in the dilated capillaries of *Vhl* cKO mice compared with those in the controls. We observed increased proliferation of ECs in the affected capillaries of *Vhl* cKO mice relative to controls, particularly within the papillary muscles and interventricular septum (Figure 3E), suggesting that local Hif activation may promote compensatory angiogenesis in response to ongoing myocardial injury. Consistent with these findings, Ingenuity Pathway Analysis indicated the enrichment of pathways related to angiogenesis and vasculogenesis (Figure S12 and Figure 3F).

Together, these results demonstrate that the loss of *Vhl* in the *Wt1* lineage leads to cardiomyocyte damage, capillary hemorrhage, interstitial fibrosis, and increased epithelial–mesenchymal transition and ECM remodeling.

Stabilization of HIF Signaling Triggers Cardiac Inflammation and Vascular Damage

Histologic analysis revealed prominent infiltration of inflammatory cells, particularly in the fibrotic regions near the epicardium, surrounding dilated or stenotic coronary vessels, and within the altered papillary muscles of the *Vhl* cKO hearts compared with the controls (Figure 4A). To better understand the transcriptional landscape associated with immune infiltration in *Vhl* cKO hearts, we revisited the functional analysis, which showed enrichment of biological processes related to cell movement and migration, chemotaxis, and innate immune responses, including those mediated by macrophages and neutrophils (Figure 4B).

To characterize the infiltrating immune cells, we performed immunofluorescence staining using the macrophage marker CD68 and the neutrophil marker MRP14 (migration inhibitory factor-related protein 14; S-100A9),

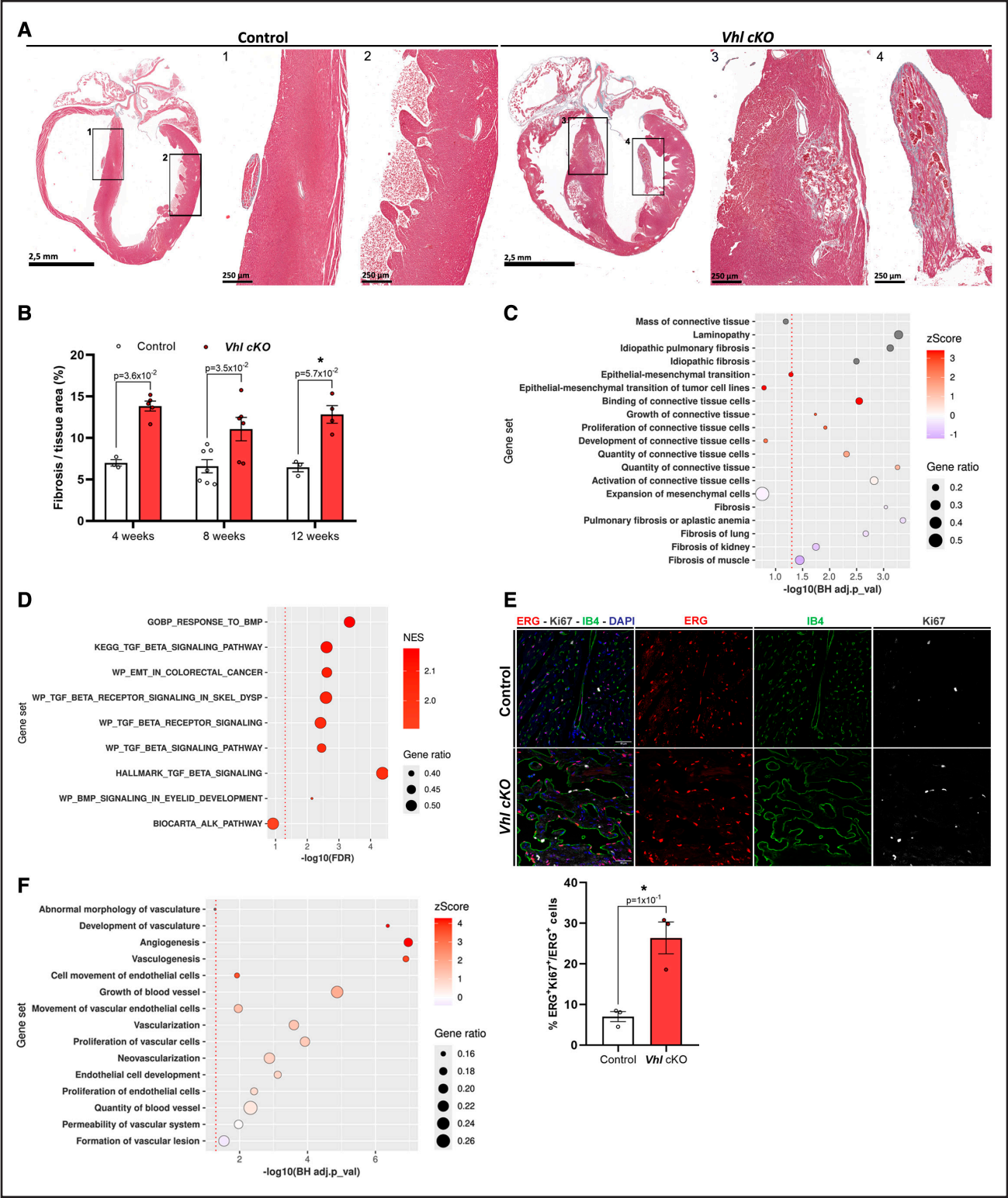


Figure 3. Elimination of *Vhl* in the *Wt1* lineage results in cardiac damage, hemorrhagic capillaries, and fibrosis.

A, Histologic analysis by Masson trichrome staining showing a representative frontal overview of cardiac sections from 8-week-old control and *Vhl* conditional knockout (cKO) mice. Scale bars, 2.5 mm. The middle (1, 3) and right (2, 4) panels show magnified views of the interventricular septum (1, 3), left ventricle (2), and papillary muscles (4), respectively. Scale bars, 250 μ m. **B**, Quantification of the fibrotic area over time determined from Masson trichrome staining of cardiac sections from control (white bars) and *Vhl* cKO mice (red bars) and expressed as percentage of tissular fibrosis. Control: 4 weeks (n=3), 8 weeks (n=7), and 12 weeks (n=3). *Vhl* cKO: 4 weeks (n=5), 8 weeks (n=6), and 12 weeks (n=4). **C**, Dot plot of selected diseases and biofunctions detected as enriched by Ingenuity Pathway Analysis in the set of differentially expressed genes associated with the contrast between *Vhl* cKO and control, representing terms related to fibrosis, (*Continued*)

Figure 3 Continued. abnormal connective tissue, and epithelial–mesenchymal transition. **D**, Dot plot of selected results from gene set enrichment analysis in the contrast of *Vhl* cKO versus control against several MSigDB gene set collections (GOBP, KEGG, WP, Hallmark and BioCarta). The dot plot presents the results for terms related to the TGF- β (transforming growth factor β) and BMP (bone morphogenetic protein) signaling pathways. Enrichment significance is represented on the x axis as $-\log_{10}$ (false discovery rate). The dot size represents the percentage of leading-edge genes relative to the gene set size, and the dot color represents the normalized enrichment score; positive values indicate that genes in the gene set have a tendency to be more expressed in *Vhl* cKO conditions. **E**, Immunofluorescence for ERG (ETS transcription factor; red), Ki67 (white), IB4 (isolectin B4; green), and DAPI (blue) in cardiac ventricular sections from control and *Vhl* cKO mice by 8 weeks of age. Scale bars, 40 μ m. Quantification of the percentage of proliferative endothelial cells (ERG⁺–Ki67⁺/ERG⁺) in the affected capillaries in the papillary muscle area and interventricular septum in *Vhl* cKO (red bar) and control mice (white bar) is shown (n=3). **F**, Dot plot of selected diseases and biofunctions obtained from Ingenuity Pathway Analysis in the set of differentially expressed genes associated with the contrast of *Vhl* cKO versus control mice for terms related to “angiogenesis” and “vasculogenesis.” Bars represent mean $\pm$ SEM. Statistical significance was evaluated using the nonparametric Mann-Whitney *U* test. The asterisks in **B** and **E** indicate that 0.0571 and 0.100 are the lowest attainable *P* values given the sample size. At least 3 independent animals per genotype from separate litters were used for each analysis. Enrichment significance is represented on the x axis as $-\log_{10}$ (Benjamini-Hochberg P_{adjusted}) in panels **C** and **F**. The dot color represents the z score (positive values indicate higher activity in *Vhl* cKO conditions) and the dot size represents the gene ratio (fraction of function-associated genes present in the focus gene set). GOBP indicates Gene Ontology Biological Processes; KEGG, Kyoto Encyclopedia of Genes and Genomes; *Vhl*, von Hippel–Lindau; WP, WikiPathways; and *Wt1*, Wilms tumor 1.

together with smooth muscle actin (SMA) to delineate the coronary vessels and the fibrotic areas. This analysis revealed extensive infiltration of CD68⁺ macrophages within the fibrotic areas of the epicardium, coronary arteries, and papillary muscles in *Vhl* cKO mice compared with that in the controls (Figure 4C). In parallel, we observed significant clusters of MRP14⁺ neutrophils at similar locations, contrasting with the lack of inflammation in the control hearts (Figure 4D). In addition, we detected CD3⁺ T cells in the perivascular areas surrounding the remodeled arteries (data not shown).

Dilated arteries from *Vhl* cKO mice exhibiting medial layer disruption showed elevated expression of MPO (myeloperoxidase), an enzyme released into the extracellular space during neutrophil degranulation, which may contribute to vascular injury and the observed discontinuities in SMA staining in *Vhl* cKO versus controls (Figure 4E). TUNEL staining confirmed increased apoptosis in regions with disrupted SMA staining in *Vhl* cKO arteries versus controls, suggesting that MPO-associated neutrophil activity in the medial wall could contribute to vascular damage and remodeling (Figure 4F).

RNA-seq data supported these findings, showing the upregulation of genes involved in chemotaxis, leukocyte adhesion and migration, and cytokine signaling. These included *Anxa1*, *Aire*, *Ackr2*, *C5ar1*, *Itgax*, *Cmklr1*, *Csf1*, and *Lgals3*, among others (Tables S1 and S2). Furthermore, reverse transcription polymerase chain reaction analysis confirmed increased expression of *Il-1 β* and chemokines such as *Cxcl1* and *Mcp1/Ccl2* in *Vhl* cKO hearts (Figure 4G).

To further evaluate systemic immune activation, we performed a hematologic analysis of whole peripheral blood from 4- and 8-week-old control and *Vhl* cKO mice. Total white blood cell counts were significantly elevated in 8-week-old *Vhl* cKO mice compared with controls, with a progressive increase from 4 to 8 weeks of age (Figure S13A). Similar trends were observed for circulating lymphocytes (Figure S13B) and monocytes (Figure S13C). A marked increase in neutrophil counts was observed in

8-week-old *Vhl* cKO mice relative to the controls (Figure S13D), correlating with the observed myocardial neutrophil infiltration (Figure 4D and 4E). Likewise, both eosinophil (Figure S13E) and basophil (Figure S13F) counts were elevated in 8-week-old *Vhl* cKO mice compared with those in the controls.

These findings collectively demonstrate that deletion of *Vhl* in the *Wt1* lineage leads to substantial cardiac inflammation, characterized by robust infiltration of macrophages and neutrophils. These results further support the role of HIF stabilization in driving both local and systemic inflammatory responses in this model.

Increased Thrombosis, Aneurysm Formation, Vascular Calcification, and Vessel Rupture in *Vhl*-Deficient Coronary Arteries

Further histologic analysis revealed a high incidence of intracoronary thrombus formation in the coronary arteries of *Vhl* cKO mice compared with that in the control hearts (Figure 5A). Occlusive large thrombi in *Vhl* cKO mice showed signs of nascent blood vessel formation (Figure 5B), likely as a compensatory response to impaired coronary perfusion. In addition, hemosiderin deposits were observed within the thrombi (Figure 5B). This phenomenon, known as recanalization, occurs in human vascular pathology, such as atherosclerosis and KD.

To investigate the molecular mechanisms underlying this prothrombotic phenotype, we analyzed the RNA-seq data set and identified significant alterations in the expression of genes associated with coagulation in *Vhl* cKO mice compared with that in the controls (Figure 5C). *Vhl* cKO mice exhibited dysregulated expression of genes encoding proteins involved in the coagulation cascade, platelet aggregation, and inflammation, such as *A2m*, *F2r*, *F2r11*, *Itga2*, *Itga2b*, and *Itgb3*, among others (Tables S1 and S2).

In addition, several serpins, which function as inhibitors of serine proteases involved in coagulation and inflammation, were also upregulated in *Vhl* cKO mice compared with controls (Table S1), including *Serpine1*

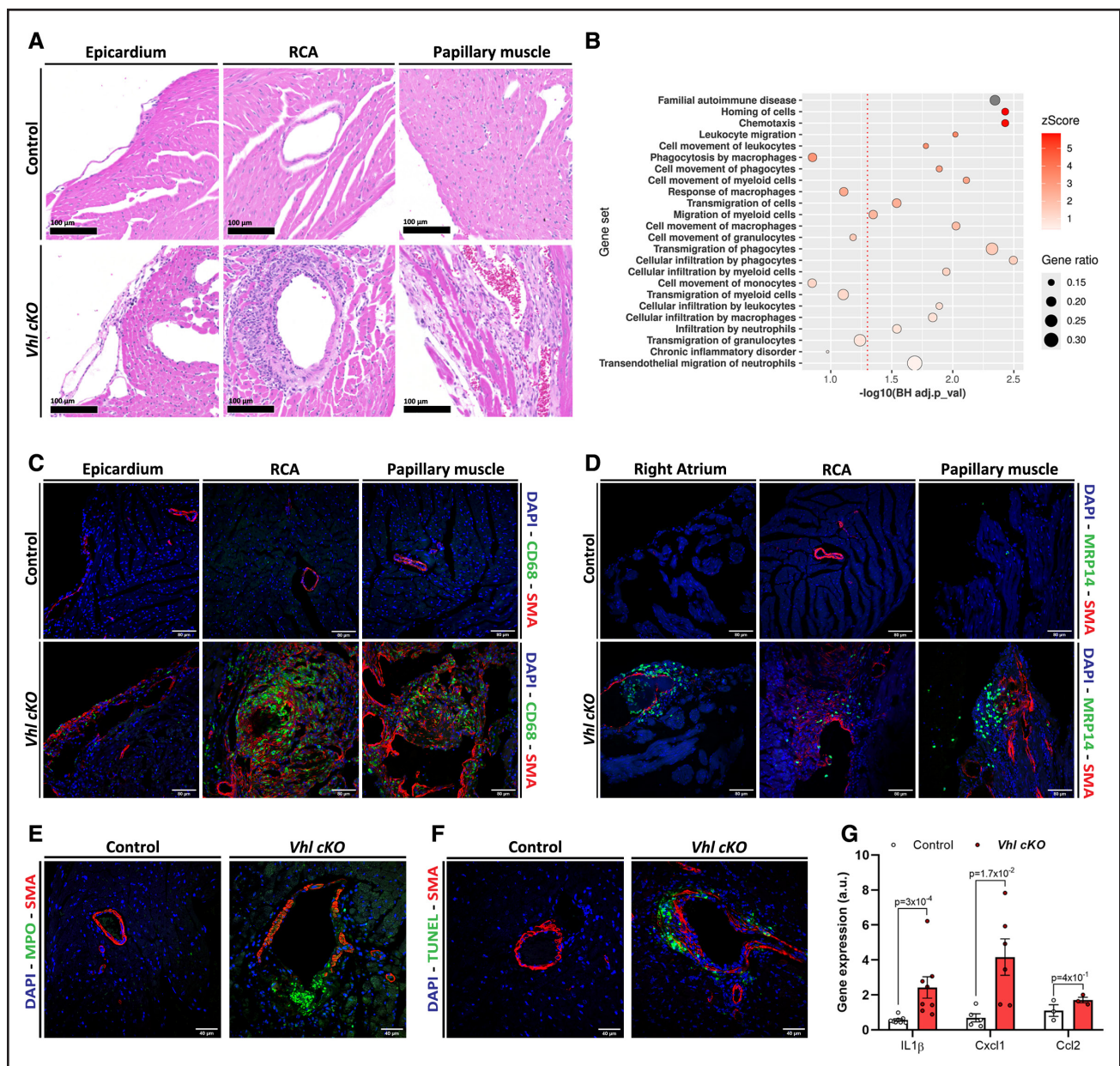


Figure 4. Stabilization of Hif signaling promotes cardiac inflammation and causes vascular damage.

A, Histologic analysis by hematoxylin & eosin staining in 8-week-old control and *Vhl* conditional knockout (cKO) hearts. Representative views of the right ventricular epicardium (left), right coronary artery (RCA, middle) and papillary muscles (right). Scale bars, 100 μ m. **B**, Dot plot of selected diseases and biofunctions obtained from Ingenuity Pathway Analysis in the set of differentially expressed genes associated with the contrast of *Vhl* cKO versus control mice representing terms related to inflammation. Enrichment significance is represented on the x axis as $-\log_{10}$ (Benjamini-Hochberg P_{adjusted}). The dot color represents the z score (positive values indicate higher activity in *Vhl* cKO mice). The dot size represents the gene ratio (fraction of function-associated genes present in the focus gene set).

C, Immunofluorescence of macrophages in 8-week-old control and *Vhl* cKO hearts by immunostaining with CD68 (green), SMA (smooth muscle actin; red), and DAPI (blue) in the same regions indicated in **A**. Scale bars, 80 μ m. **D**, Immunofluorescence of neutrophils labeled with MRP14 (migration inhibitory factor-related protein 14; green), SMA (red), and DAPI (blue) in 8-week-old control and *Vhl* cKO mice in the same regions shown in panel **A**. Scale bars, 80 μ m. **E**, MPO (myeloperoxidase) protein expression around coronary vessels was assessed by immunofluorescence in ventricular cardiac sections from control and *Vhl* cKO mice at 8 weeks. MPO (green), SMA (red), and DAPI (blue). Scale bars, 40 μ m. **F**, Apoptosis evaluation by TUNEL staining (green) in ventricular cardiac sections from 8-week-old control and *Vhl* cKO mice. Scale bars, 40 μ m. **G**, Relative gene expression analysis of the inflammatory mediators *Il1 β* , *Cxcl1*, and *Ccl2* by reverse transcription quantitative polymerase chain reaction from heart lysates of 8-week-old control (white bars) and *Vhl* cKO (red bars) mice. Control: *Il1 β* (n=8), *Cxcl1* (n=5), *Ccl2* (n=3). *Vhl* cKO: *Il1 β* (n=8), *Cxcl1* (n=6), *Ccl2* (n=3). Bars represent mean $\pm$ SEM. Statistical significance was achieved using the nonparametric Mann-Whitney *U* test. At least 3 independent animals per genotype from separate litters were used for each analysis. HIF indicates hypoxia-inducible factor; and Vhl, von Hippel-Lindau.

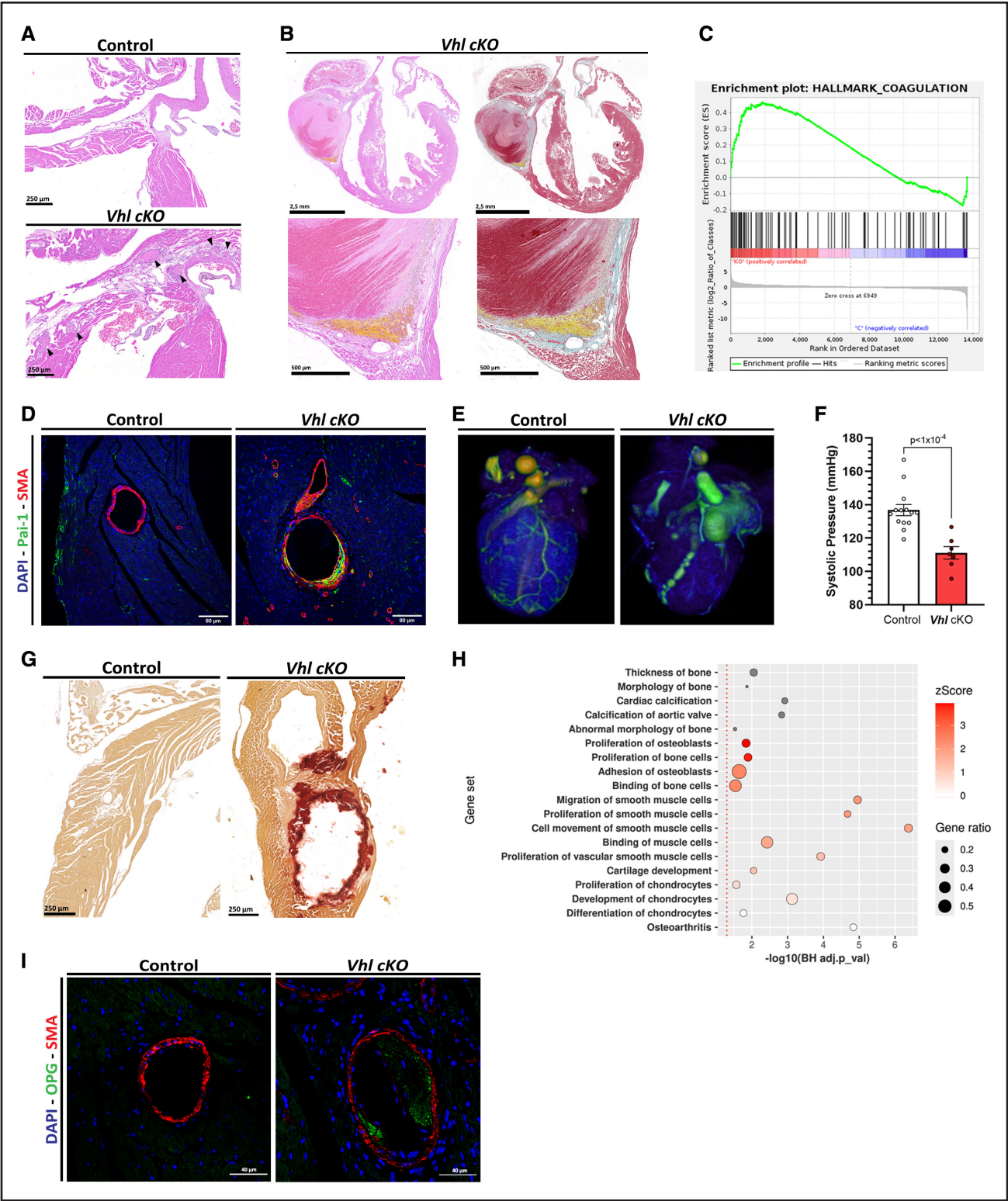


Figure 5. Increased thrombus formation, vascular calcification, and vessel rupture in *Vhl*-deficient coronary arteries. **A**, Histologic analysis by hematoxylin & eosin staining of heart sections from 8-week-old control and *Vhl* conditional knockout (cKO) mice showing a magnified region of the right ventricle and atrium. The arrowheads in the *Vhl* cKO mice indicate thrombi in the coronary arteries. Scale bars, 250 μ m. **B**, Histologic analysis by hematoxylin & eosin (left panels) and Masson trichrome (right panels) staining in frontal heart sections of 8-week-old *Vhl* cKO mice highlighting a large intracoronary thrombus near the right coronary artery (top panels). Scale bars, 2.5 mm. Magnifications of the thrombus (bottom panels) show fibrin (pink) and collagen (blue) deposition, hemosiderin (yellow) deposits, and nascent blood vessels. Scale bars, 500 μ m. **C**, Gene set enrichment analysis enrichment plot for the Hallmark database coagulation gene set in the contrast of *Vhl* cKO versus control mice. The red to blue stripe represents 13 677 genes detected as expressed after differential expression analysis, ranked by logFC. (Continued)

Figure 5 Continued. Genes on the left side (red) were highly expressed in the *Vhl* cKO samples, and those located on the right side (blue) were highly expressed in the control samples. Vertical black lines represent the positions of genes in the coagulation gene set along the ranked collection of genes. The green curve represents the cumulative enrichment score. **D**, Representative immunofluorescence staining for Pai-1 (plasminogen activator inhibitor-1; green), SMA (smooth muscle actin; red), and DAPI (blue) in ventricular coronary arteries from 8-week-old control and *Vhl* cKO mice. Scale bars, 80 μ m. **E**, Coronary architecture evaluation by Microfil resin-cast and computed tomography (CT) imaging analysis of 8-week-old control and *Vhl* cKO hearts showing aneurysm-like vascular lesions in the coronary arteries of *Vhl* cKO mice. **F**, Values of systolic pressure registered in the control (white bar; n=14) and *Vhl* cKO (red bar; n=7) mice at 8 weeks of age. **G**, Representative Alizarin Red staining of 8-week-old control and variant (*Vhl* cKO) heart sections at the right ventricular coronary artery level identifying calcium deposits (red). Scale bars, 250 μ m. **H**, Dot plot of selected diseases and biofunctions obtained from Ingenuity Pathway Analysis in the set of differentially expressed genes associated with the contrast of *Vhl* cKO versus control mice for terms related to "calcification of blood vessels," "osteogenesis," "chondrogenesis," and "smooth muscle cell proliferation, migration and calcification." **I**, Immunofluorescence staining for OPG (osteoprotegerin; green), SMA (red), and DAPI (blue) in coronary arteries from 8-week-old control and *Vhl* cKO mice. Scale bars, 40 μ m. Bars represent mean $\pm$ SEM. Statistical significance was tested using the nonparametric Mann-Whitney *U* test. Enrichment or downregulation significance is represented on the x axis as $-\log_{10}$ (Benjamini-Hochberg P_{adjusted}) in **H**. The dot color represents the z score (positive values indicate higher activity in *Vhl* cKO mice) and the dot size represents the gene ratio (fraction of function-associated genes present in the focus gene set). At least 3 independent animals per genotype from separate litters were used for each histologic analysis. *Vhl* indicates von Hippel-Lindau.

(Pai-1 [plasminogen activator inhibitor 1]), a major inhibitor of tissue-type plasminogen activator and urokinase-type plasminogen activator and a well-established HIF target gene.^{34,35} Immunofluorescence analysis confirmed significantly increased Pai-1 protein expression in the coronary arteries of *Vhl* cKO mice relative to controls (Figure 5D), suggesting that impaired fibrinolysis due to reduced plasminogen conversion into plasmin may contribute to the high incidence of thrombi formation observed in *Vhl* cKO mice.

To further characterize the vascular abnormalities, we performed 3-dimensional computed tomography imaging of *Vhl* cKO hearts at 8 weeks of age after coronary resin cast filling. The analysis revealed aneurysm-like vascular lesions in the coronary arteries, which were absent in the control mice (Figure 5E). These lesions progressed to coronary ectasia over time (data not shown). Functional analysis showed enrichment of gene categories related to vasodilation and arterial occlusion, although genes typically associated with aortic aneurysms were downregulated, in agreement with the lack of aortic involvement in the *Vhl* cKO mice (Figure S14A and Table S2).

Given the link between vasodilation and reduced vascular resistance, we assessed the systemic arterial pressure. Systolic blood pressure was significantly reduced in *Vhl* cKO mice relative to that in the controls (Figure 5F), correlating with increased expression of vasodilators such as *Adm* (adrenomedullin; *Adm*), *Apln* (apelin; *Apln*), and *Bdkrb2* (bradykinin receptor B2; *Bdkrb2*; Figure S14B and Tables S1 and S2).

In addition, vascular calcification was detected by Von Kossa (Figure S14C) and Alizarin Red staining (Figure 5G) in the most dilated, remodeled arteries of *Vhl* cKO hearts, but not in controls, suggesting that increased vessel rigidity and stiffness may contribute to coronary rupture, pericardial hemorrhage, and sudden death in this model. Arterial calcification involves the differentiation of VSMCs into chondrocyte- or osteoblast-like cells, a process driven by inflammation and elevated calcium levels.³⁶ Functional analysis revealed the upregulation of

genes associated with vascular calcification, osteogenesis, chondrogenesis, and smooth muscle cell proliferation and migration (Figure 5H and Table S2), suggesting that activation of HIF signaling leads to the dedifferentiation of VSMCs into a chondrocyte- or osteoblast-like phenotype in *Vhl* cKO hearts. Among the altered expressed genes, *Tnfrsf11b*, which encodes OPG (osteoprotegerin), was significantly upregulated in *Vhl* cKO hearts compared with controls (Table S1). Furthermore, substantial elevation of OPG was observed in the remodeled arteries from *Vhl* cKO mice (Figure 5I). Despite its function as a RANKL (receptor activator of nuclear factor κ B ligand) decoy receptor, OPG has been linked to increased risk of CVD and endothelial damage in humans,³⁷ and a high OPG serum level has been proposed as a biomarker of KD with arterial involvement.³⁸

Overall, deletion of *Vhl* in Wt1 lineages activates a gene expression program that promotes thrombosis, platelet aggregation, arterial dilation, vascular calcification, and vessel rupture, mimicking pathophysiologic features of atherosclerosis and KD. These findings highlight the vital role of VHL in maintaining coronary vessel integrity by regulating coagulation, ECM remodeling, and VSMC homeostasis.

Patients With KD Display Activation of HIFs Around Severe Coronary Vascular Lesions

Analysis of the *Vhl* cKO mouse model indicated that activation of Hif signaling in the Wt1 lineage leads to cardiovascular abnormalities that are hallmark complications of the most severe cases of KD.^{22,23}

Histologic analysis of ventricular transverse sections from a post-mortem case of a 3-month-old female infant who had KD revealed marked coronary remodeling with a remarkable aneurysm and thrombus at the level of the left main coronary artery as evidenced by hematoxylin & eosin staining (Figure 6A). Extensive immune infiltration was observed in the remodeled occluded artery (Figure 6B). In contrast, remote regions of the left ventricular myocardium, distant from the affected vessel, appeared

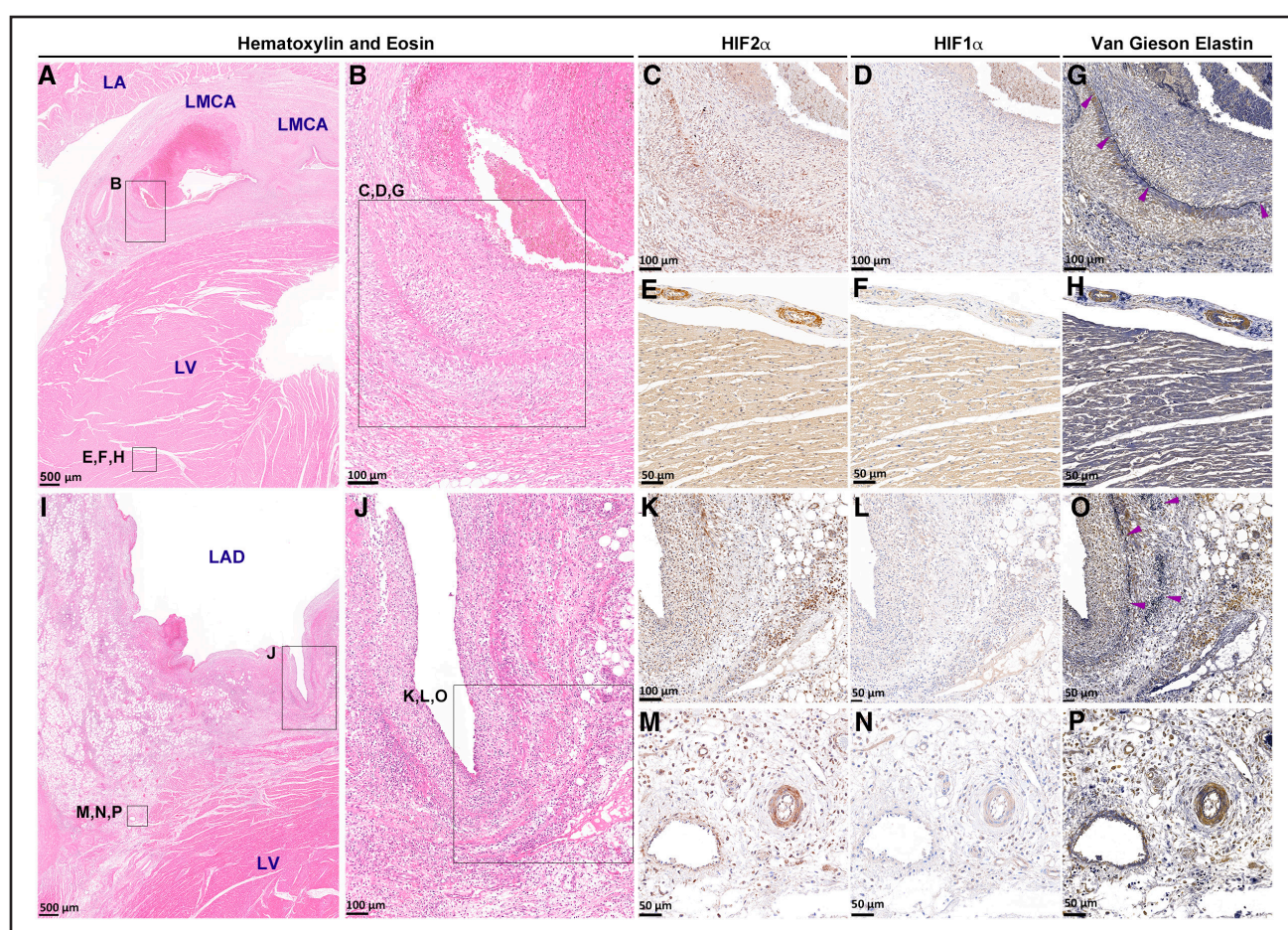


Figure 6. Activation of HIFs around severe coronary vascular lesions in patients with Kawasaki disease.

A through **H**, Histologic analysis of representative cardiac sections from a post-mortem case of a 3-month-old female infant who had Kawasaki disease. **A**, Hematoxylin & eosin staining showing vascular remodeling and thrombus formation in the left main coronary artery (LMCA; top inset; **B**) and healthy myocardium of the left ventricle (LV) without signs of inflammation or vascular lesions (bottom insets in **E**, **F**, and **H**). Scale bar, 500 μ m. **B**, Zoom of the LMCA with high vascular remodeling observed by hematoxylin & eosin staining. Insets **C**, **D**, and **G** highlight the inflammatory area around the coronary vessel. Scale bar, 100 μ m. **C** through **F**, Immunohistochemistry staining of HIF2 α (hypoxia-inducible factor 2 α ; **C** and **E**) and HIF1 α (hypoxia-inducible factor 1 α ; **D** and **F**) in the area of coronary remodeling (**C** and **D**; scale bars, 100 μ m) and healthy myocardium (**E** and **F**; scale bars, 50 μ m). The nuclei were counterstained with hematoxylin. **G** and **H**, Verhoeff–Van Gieson elastin staining on top of HIF2 α staining showing elastin breaks (purple arrowheads) in the remodeled artery (**G**; scale bar, 100 μ m) and normal internal lamina in non-remodeled vessels (**H**; scale bar, 50 μ m). **I** through **P**, Independent histologic cardiac sections from a post-mortem case of an 8-month-old female infant who had Kawasaki disease. **I**, Hematoxylin & eosin staining showing profound coronary remodeling of a giant aneurysm in the left anterior descending (LAD) coronary artery without thrombus (top inset; **J**) and conserved left ventricular myocardium (bottom insets; **M**, **N**, and **P**). Scale bar, 500 μ m. **J**, Zoom of the LAD vascular remodeling stained with hematoxylin & eosin. Insets **K**, **L**, and **O** highlight an inflammatory area around the LAD coronary artery. Scale bar, 100 μ m. **K** through **N**, Immunohistochemistry of HIF2 α (**K** and **M**) and HIF1 α (**L** and **N**) in the epicardial area with remarkable coronary remodeling (**K**; scale bar, 100 μ m; **L**; scale bar, 50 μ m) and around a partially remodeled vessel in the ventricular myocardium (**M** and **N**; scale bars, 50 μ m). The nuclei were counterstained with hematoxylin. **O** and **P**, Verhoeff–Van Gieson elastin staining showing elastin breaks (purple arrowheads) in the highly remodeled LAD coronary artery (**O**; scale bar, 50 μ m) and normal internal lamina in a nonremodeled vessel (**P**; scale bar, 50 μ m). Two independent cases from the 5 analyzed are shown. LA indicates left atria.

structurally intact (Figure 6A, bottom insets, and 6E, 6F, and 6H).

Immunohistochemistry revealed robust expression of HIF2 α in the medial layer of the remodeled vessels and inflammatory cells surrounding the lesions (Figure 6C and Figure S15A). In comparison, cytoplasmic HIF2 α was weakly detected in the coronary arteries from unaffected areas but not in the adjacent myocardium (Figure 6E). In contrast, HIF1 α expression was limited, with only occasional staining in a few inflammatory cells near

the lesion and thrombus, and was weakly expressed in the media or intima of both diseased and healthy arterial regions (Figure 6D and 6F). Furthermore, elastin fragmentation and discontinuity were confirmed in the remodeled arteries (Figure 6G), whereas the elastic lamina remained intact in the unaffected coronary vessels (Figure 6H).

In a second post-mortem case from an 8-month-old female infant who had KD (Figure 6I and 6J), we observed similar patterns: HIF2 α was detected in the

medial and intima layers, and prominently expressed in macrophages in and around the remodeled left anterior descending coronary artery, where a giant aneurysm had formed (Figure 6K). Consistent with the former case, HIF1 α was barely detected in these regions, including inflammatory infiltrates (Figure 6L and Figure S15B). In partially affected coronaries located near the histologically preserved myocardium (Figure 6I, bottom panels, and 6M, 6N, and 6P), HIF2 α was again strongly expressed in the inflammatory infiltrate around the vessels (Figure 6M), whereas HIF1 α remained mostly undetectable (Figure 6N). In this patient, elastin breaks were evident in the highly remodeled arteries (Figure 6O), whereas the medial integrity and elastic lamina remained intact in the unaffected coronary vessels (Figure 6P).

A comparable expression pattern of strong HIF2 α and minimal or absent HIF1 α signals was observed in 3 additional KD post-mortem cases (male infants aged 7 months to 3 years, who experienced fatal cardiac events within 10 to 30 days after the acute KD phase; data not shown).

These observations collectively demonstrate that the hypoxia signaling pathway, particularly HIF2 α activation, is consistently enriched in areas of coronary vascular remodeling and inflammation in patients with KD and severe cardiac involvement. These data suggest that HIF2 signaling may play an active and previously underappreciated role in promoting arterial remodeling and immune activation during the acute phase of KD.

Elimination of *Hif2* Rescues Cardiomegaly, Fibrosis, Inflammation, and Coronary Abnormalities in the *Vhl*/Wt1 Mouse Model

Given the marked HIF2 α upregulation in the coronary lesions of both *Vhl* cKO mice and human patients with KD, we generated a double knockout (DKO) mouse model lacking both *Vhl* and *Hif2a* in the Wt1 lineage (*Vhl*/*Hif2a*/Wt1 DKO) to evaluate the specific contribution of Hif2 signaling to the cardiac abnormalities observed upon *Vhl* deletion.

Eight-week-old DKO mice exhibited a qualitative rescue of cardiac size and morphology relative to *Vhl* cKO mice (Figure 7A), with a heart weight–body weight ratio comparable to that of the controls (Figure 7B). Echocardiography confirmed normalized chamber dimensions (data not shown), and restoration of left ventricular mass (Figure 7C) and coronary artery diameter (Figure 7D), in contrast to the cardiomegaly and coronary dilation evident in the *Vhl* cKO mice.

Histologic analysis by hematoxylin & eosin and Masson trichrome staining further demonstrated that DKO hearts lacked the extensive vascular remodeling and thrombi formation observed in *Vhl* cKO mice (Figure 7E), displayed preserved capillary structure in the papillary muscles (Figure 7F), and showed no signs

of atrial or epicardial fibrosis (Figure 7G). The marked macrophage and neutrophil infiltration that characterized the inflamed regions of the single *Vhl* mutants (Figure 4) was absent in DKO hearts (Figure 7H), indicating that the loss of *Hif2a* in Wt1-derived cells is sufficient to prevent the inflammatory response triggered by *Vhl* deficiency.

To investigate the molecular basis of this phenotypic rescue, we performed bulk RNA-seq of ventricular tissues from DKO and *Vhl* cKO mice. The transcriptomic profile of DKO samples clustered closely with controls (Figure S16), with only 7 differentially expressed genes between DKO and control hearts, compared with 1692 differentially expressed genes between *Vhl* cKO and DKO samples (Figure 7I and Table S3). These findings strongly suggest that HIF2 is a major driver of the transcriptional dysregulation observed in *Vhl*-deficient hearts, modulating gene networks involved in ECM remodeling, angiogenesis, VSMC behavior, inflammation, thrombosis, and vascular calcification.

Taken together, our findings demonstrate that the constitutive activation of Hif2 signaling in Wt1 lineage cells after *Vhl* deletion is sufficient to drive the major features of cardiac defects observed in this model. The data identified Hif2 as a key molecular effector of coronary vessel remodeling, vascular inflammation, thrombosis, and mineralization (Figure 8). These alterations mirror the main cardiovascular manifestations observed in severe cases of KD, providing a mechanistic link between hypoxia signaling and KD pathogenesis. Overall, our results highlight HIF2 as a central mediator of coronary pathology and a promising therapeutic target for the treatment of KD and other disorders involving aberrant vascular remodeling.

DISCUSSION

In this study, we investigated the role of VHL and HIF signaling in coronary vessel development and homeostasis by characterizing novel genetic mouse models targeting the *Vhl*/Hif axis in Wt1-derived lineages. We showed that deletion of *Vhl* in Wt1-expressing cells does not preclude normal coronary vessel development per se, but leads to severe vascular pathology in adult mice, including cardiomegaly and cardiac fibrosis, with preserved cardiac function. *Vhl* cKO mice progressively develop coronary artery dilation accompanied by disrupted ECM organization, stabilization of HIF proteins, and increased activity or expression of Mmp2, Mmp9, Timp1, Postn, Pai-1, Bmp2, Acan, and Mpo. These alterations are associated with arterial remodeling, vascular damage, loss of VSMCs, and macrophage- and neutrophil-mediated vasculitis. Furthermore, *Vhl* cKO mice displayed a high incidence of intracoronary thrombi and vascular calcification. Together with excessive collagen and fibronectin deposition and increased osteoprotegerin expression at sites of vascular inflammation, these differences may enhance

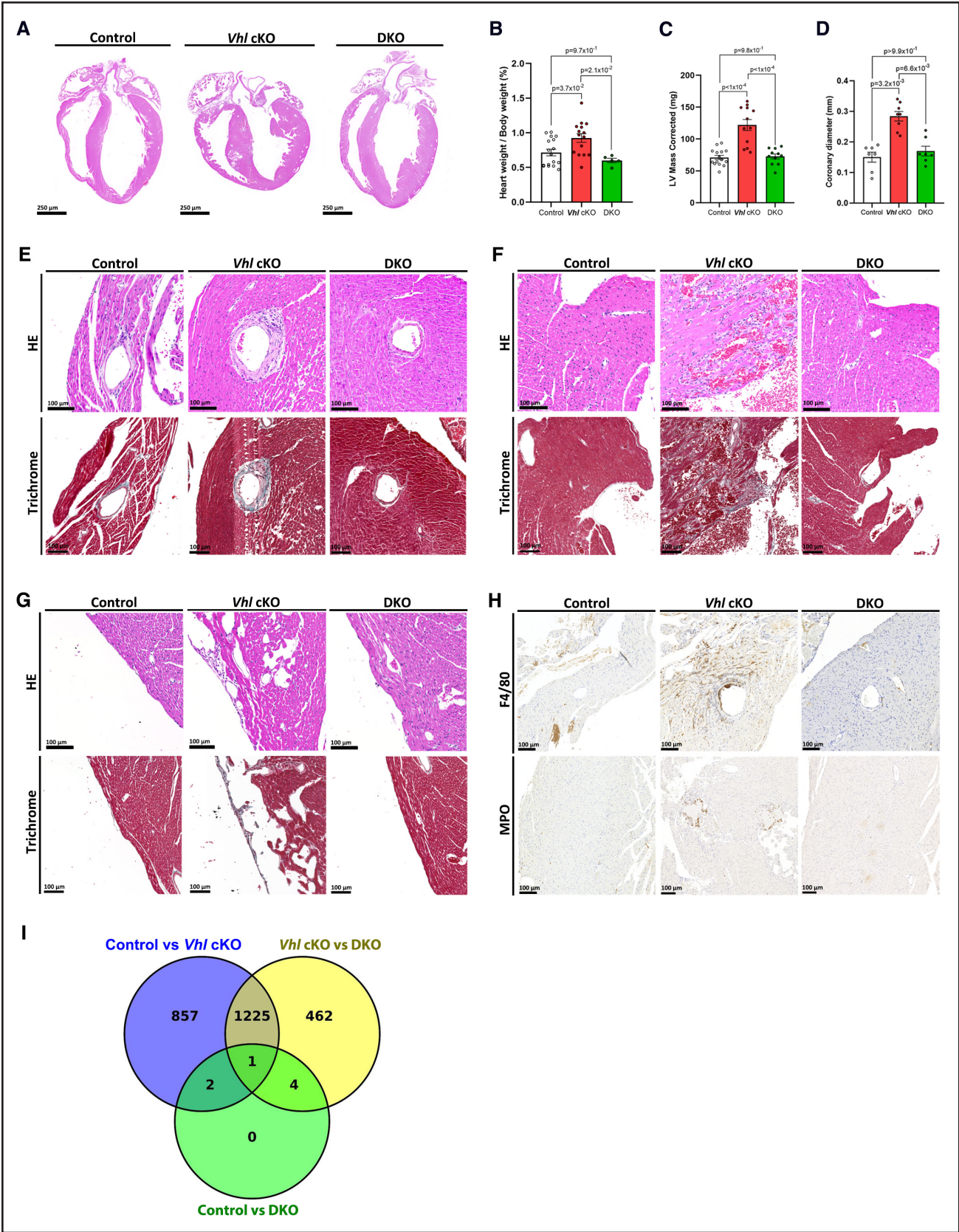


Figure 7. Elimination of *Hif2* rescues cardiomegaly, fibrosis, inflammation, and coronary abnormalities in the *Vhl*/*Wt1* mouse model.
A, Histologic analysis of heart structure in a 4-chamber view by hematoxylin & eosin staining of 8-week-old control, *Vhl* conditional knockout (cKO), and double *Vhl*/*Hif2a* knockout (DKO) mouse cardiac sections. Scale bar, 250 μ m. **B**, Cardiac size estimated as heart weight–body (*Continued*)

Figure 7 Continued. weight ratio of the indicated mice: control (white bars; n=17), *Vhl* cKO (red bars; n=15), and DKO (green bars; n=5). **C** and **D**, Echocardiographic analysis in 8-week-old mice showing the left ventricular corrected mass (**C**) and the coronary diameter (**D**) in the indicated model: control (n=15–7), *Vhl* cKO (n=12–10), and DKO (n=11–7). **E** through **G**, Histologic analysis by hematoxylin & eosin (top panels) and Masson trichrome staining (bottom panels) showing the right ventricular coronary artery structure (**E**), architecture of the interventricular septum and papillary muscles (**F**), and right ventricular epicardium integrity (**G**) in 8-week-old control, *Vhl* cKO, and DKO mice. Scale bars, 100 μ m. **H**, Representative immunohistochemistry of the inflammatory infiltrate by staining with F4/80 (top panels) and MPO (myeloperoxidase; bottom panels) in the right ventricle of the indicated mouse group at 8 weeks of age. Scale bars, 100 μ m. **I**, Venn diagram showing shared and specific differentially expressed genes detected in 8-week-old mice in the contrasts control versus *Vhl* cKO, *Vhl* cKO versus DKO, and control versus DKO. Bars represent mean $\pm$ SEM. Statistical significance was assessed using the nonparametric Kruskal-Wallis test followed by the Dunn multiple comparisons test (**B** and **D**) and the Shapiro-Wilk test, 1-way ANOVA followed by Tukey multiple comparisons test (**C**). At least 3 independent animals per genotype from separate litters were used for each analysis. Hif2 indicates hypoxia-inducible factor 2; LV, left ventricle; *Vhl*, von Hippel Lindau; and Wt1, Wilms tumor 1.

vascular stiffness and predispose vessels to rupture, potentially contributing to the increased mortality observed in adult *Vhl* cKO mice (Figure 8).

These alterations in *Vhl* cKO mice closely recapitulate the key cardiac complications reported in patients with severe KD, a rare pediatric vasculitis representing the leading cause of acquired heart disease in children. Here, we identified marked HIF2 α expression in the medial and intima layers and infiltrating immune cells within the affected coronary arteries in both male and female patients with KD with fatal outcomes.

Genetic ablation of *Hif2 α* in Wt1 lineage cells (*Vhl/Hif2 α /Wt1-Cre*; DKO) prevented vascular damage, inflammation, and remodeling; restored coronary structure and cardiac morphology; and normalized transcriptional alterations induced by *Vhl* loss. These results underscore the essential role of HIF2 in maintaining vascular integrity and suggest a central function of this transcription factor in the pathogenesis of KD.

Deletion of *Vhl* in the Wt1 lineage leads to cardiomegaly and hypertrophy. Wt1 labels epicardial progenitors contributing to coronary VSMCs, a subset of ECs and CFs,¹⁴ and is also expressed in cardiac capillaries^{17–19} and a small proportion of embryonic CMs²⁰ (Figure S4A–S4D). Thus, an important question is whether cardiac hypertrophy in *Vhl* cKO mice arises through cell-autonomous mechanisms or is secondary to micro-environmental changes. Lineage tracing using Wt1 Td-Tomato reporter mice revealed only a sparse contribution of Wt1 to CMs, with regional enrichment in the interventricular septum and basal apex of both ventricles (Figure S8A–S8D). Furthermore, our former model of *Vhl* deletion in embryonic CMs using *Nkx2.5-Cre* did not result in cardiomyocyte hypertrophy,¹¹ suggesting that cardiac hypertrophy in the *Vhl/Wt1* cKO model likely arises through non-cell-autonomous mechanisms.

Previous studies have shown that angiogenesis can promote myocardial hypertrophy through the secretion of paracrine signals, such as VEGF-B or PR39.³⁹ Consistent with this, we observed increased EC proliferation (Figure 3) and increased expression of genes related to angiogenesis and vasculogenesis in *Vhl* cKO mice (Figure S7, Table S2, and Figure 3F). Inducible deletion of *Phd2* (an upstream negative regulator of the Hif- α subunits) in

ECs has been shown to trigger left ventricular hypertrophy and fibrosis through a Hif2-dependent mechanism.⁴⁰ Thus, activation of Hif2 signaling in the endothelium of *Vhl* cKO mice likely contributes to both cardiomegaly and fibrosis. CFs derived from Wt1 epicardial progenitors represent another potential source of paracrine cues. CFs are essential for maintaining the structural integrity of the heart by regulating ECM stability and remodeling. Upon cardiac injury, CFs differentiate into myofibroblasts with a secretory phenotype, releasing proinflammatory and profibrotic cytokines, including IL-6, IL-1 β , IL-17, TGF- β , TNF- α , and MCP-1 (monocyte chemoattractant protein-1),^{41,42} which can exacerbate cardiomyocyte hypertrophy, fibrosis, and inflammation. Many of these factors are direct HIF targets, providing a plausible mechanistic link between HIF2 activation and the observed pathologic features. Defining the specific molecular and cellular mediators of cardiac hypertrophy, fibrosis, and inflammation in the *Vhl* cKO model will require further investigation.

Regarding the vascular abnormalities observed in the *Vhl* cKO model, our data demonstrated that the constitutive activation of HIFs in the coronary arteries leads to vessel dilation, extensive vascular remodeling, and inflammation. HIFs mediate vasodilation through the upregulation of adrenomedullin, a potent vasodilatory peptide that stimulates endothelial nitric oxide production, leading to VSMC relaxation.⁴³ We consistently observed increased gene expression of *Adm* (*Adm*) and *Adm2* (*Adm2*) in *Vhl* cKO hearts (Table S1), suggesting that Hif-driven induction of these vasoactive mediators may underlie, at least in part, the observed coronary dilation. ADM levels are increased in the plasma⁴⁴ and blood monocytes⁴⁵ of patients with KD during the acute phase. Coronary dilation may also be related to ECM remodeling, as increased *Mmp9*/*Mmp2*/*Timp1* activity or expression was detected in *Vhl* cKO mice (Figure 2). *Mmp9* has been implicated in elastin breakdown in the *Lactobacillus casei* cell wall extract-induced model of KD⁴⁶ and both MMP9 and TIMP1 are elevated in the plasma of patients with KD during the acute phase.⁴⁷

In other models of Hif activation, such as inducible *Phd2* deletion in adult mice, hyperactive angiogenesis and increased capillary density in the heart are accompanied by vessel enlargement. *Phd2*-cKO mice also develop

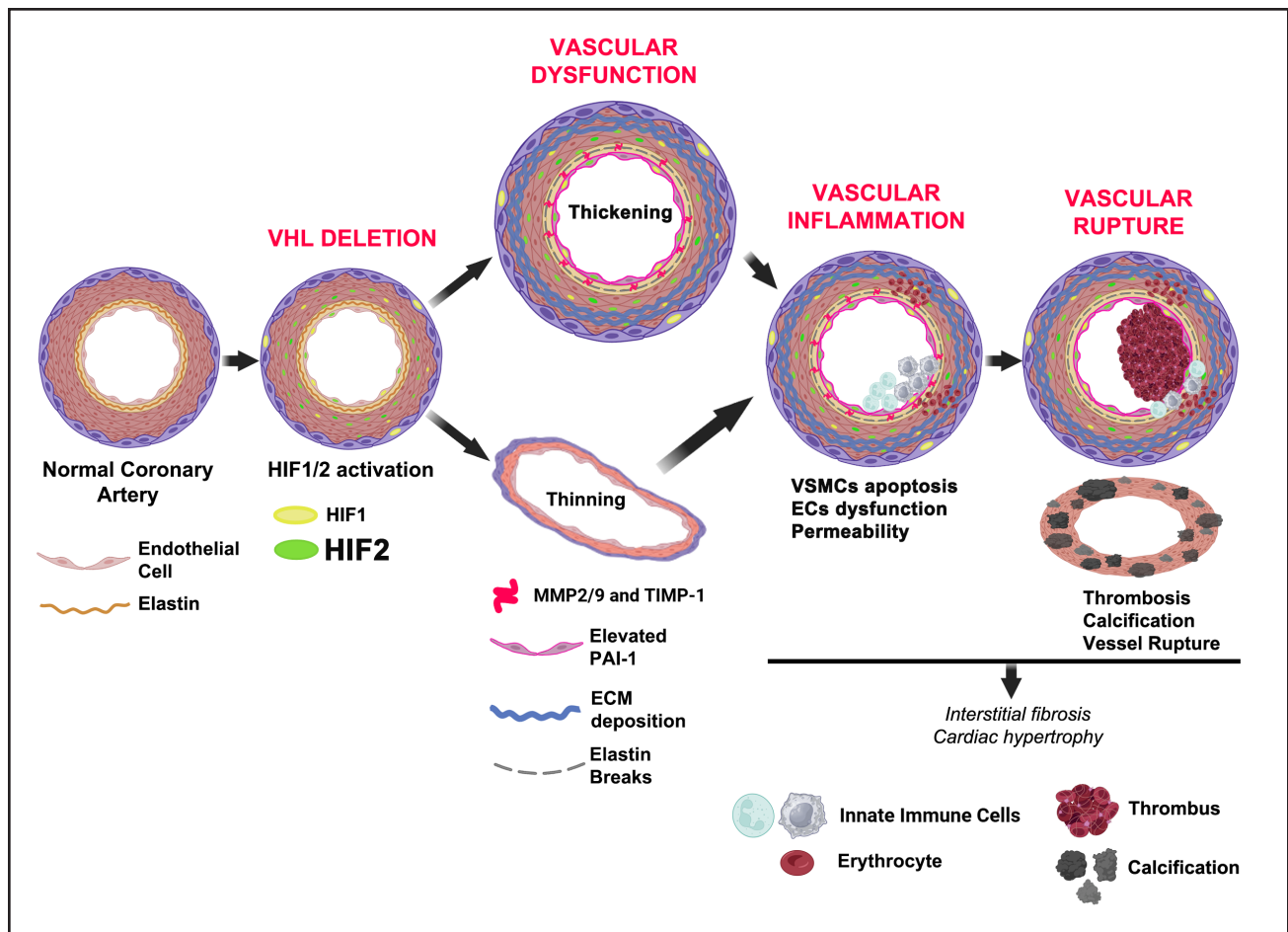


Figure 8. Proposed model of HIF-induced arterial vascular remodeling.

The proposed model of arterial vascular remodeling shows that deletion of *Vhl* in the Wt1 (Wilms tumor 1) lineage leads to activation of HIF1 (hypoxia-inducible factor 1) and mainly HIF2 (hypoxia-inducible factor 2) in the coronary arteries. This genetic deletion promotes vascular dysfunction with increased extracellular matrix (ECM) deposition (eg, FN1 [fibronectin 1], collagens, POSTN [periostin], ACAN [aggrecan]) and elevated expression of MMP2 (metalloproteinase 2), MMP9 (metalloproteinase 9), and TIMP1 (tissue inhibitor of metalloproteinase 1), together with increased levels of PAI-1 (plasminogen activator inhibitor 1) and BMP2 (bone morphogenetic factor 2), causing elastin breaks and vascular remodeling (thickening and thinning of the arterial wall). These vascular alterations lead to vascular smooth muscle cell (VSMC) apoptosis, endothelial cell (EC) dysfunction, increased permeability, and inflammatory infiltration by macrophages and neutrophils. *Vhl*-null arteries undergo profound remodeling, with a high incidence of intracoronary thrombi, calcification, and rupture. In addition, *Vhl* conditional knockout mice display interstitial fibrosis and cardiac hypertrophy, most likely through indirect paracrine signaling. HIF1 indicates hypoxia-inducible factor 1; HIF2, hypoxia-inducible factor 2; VHL, von Hippel–Lindau.

angiectasia in multiple organs, including the lungs, liver, and kidneys.⁴⁸ Both arteries and veins are enlarged in this model, and only Hif1 α levels, but not Hif2 α levels, are elevated.⁴⁸ In contrast, *Vhl*/Wt1 cKO mice display arterial but not venous dilation. These differences probably reflect the distinct outcomes of systemic (*Phd2* KO) versus lineage-restricted (*Vhl*/Wt1 cKO) Hif activation, suggesting that localized and cell type–specific induction of Hif2 may uniquely contribute to coronary arterial remodeling.

Consistent with these findings, functional analysis of *Vhl* cKO hearts revealed altered expression of genes associated with arterial aneurysm formation and rupture, as well as genes linked to syndromes characterized by aortic dilation, such as Marfan syndrome or Loeys-Dietz syndrome (Table S2 and Figure S14A). Moreover, we detected the

upregulation of multiple genes involved in endothelial activation and vascular permeability (Tables S1 and S2), reinforcing the notion that Hif2-driven signaling in Wt1-derived cells disrupts coronary vascular homeostasis.

The lack of specific diagnostic markers and targeted therapies represents a burden in reducing cardiovascular risk of coronary dilation and aneurysms in patients with KD, especially those with delayed clinical diagnosis or treatment. A deeper understanding of the molecular mechanisms underlying KD pathogenesis is essential for developing improved diagnostic tools and therapeutic strategies capable of minimizing long-term cardiac complications and irreversible sequelae. Due to the limited access to diseased human coronary tissues, the development and refinement of experimental mouse models have advanced our understanding of KD pathophysiology.

These models have helped identify key immune pathways and cellular mediators involved in vascular injury and have supported the preclinical evaluation of novel therapeutic candidates now under investigation in clinical trials.^{27,49–51}

Current animal models of KD, including those induced by the systemic administration of bacterial and fungal cell wall extracts (eg, *Lactobacillus casei* cell wall extract or *Candida albicans* water-soluble fraction), provoke widespread immune activation that affects multiple vascular beds, including coronary arteries. In addition to coronary lesions, these models often display prominent aortitis and abdominal aortic aneurysms, accompanied by systemic vascular inflammation. Patients with KD may develop severe aortitis, but rarely exhibit aortic dilation or aneurysm formation,⁵² suggesting that models primarily describing aortic aneurysm pathology may not fully capture the coronary-specific vascular damage seen in the most critical cases of KD with cardiac involvement. In this context, the *Vhl*/Wt1 cKO mouse model presented here offers an alternative and complementary tool for studying KD. This mouse model faithfully recapitulates the hallmark cardiac complications observed in severe KD, including coronary artery dilation, elastin fragmentation, thrombosis, and myocardial remodeling, without accompanying aortic dilation. This cardiac specificity enhances the model's relevance to human diseases.

We identified strong HIF2 α expression in dilated coronary arteries and inflammatory cells surrounding vascular lesions in autopsy samples from patients with KD with lethality. To our knowledge, this is the first genetic mouse model to implicate HIF signaling in KD pathogenesis. The *Lactobacillus casei* cell wall extract and *Candida albicans* water-soluble fraction models have been instrumental in modeling systemic immune-mediated vasculitis; the *Vhl*/Wt1 cKO model provides a distinct genetic platform to study intrinsic vascular mechanisms, particularly the role of HIF2 activation, which may drive coronary artery remodeling in KD.

Although HIF stabilization in our model resulted from targeted *Vhl* deletion, we do not propose that *Vhl* sequence variations underlie KD. Rather, we hypothesize that noncanonical inhibition of PHDs by inflammatory cytokines, metabolic intermediates, nitric oxide, or reactive oxygen species⁵³ may lead to localized HIF2 stabilization during the onset of the acute inflammatory phase of KD. HIF signaling has been shown to modulate immune cell infiltration, chemotaxis, and phagocytic capacity,⁵⁴ and our findings of elevated HIF2 expression in macrophages surrounding KD-affected coronary arteries suggest novel molecular pathways contributing to vascular inflammation and remodeling.

The *Vhl*/Wt1 cKO model does not aim to replace existing KD models, but to complement them by enabling a focused study of coronary-specific pathology and its progression to severe vascular complications, such as aneurysm rupture and sudden cardiac death, which are

rare but devastating outcomes in KD. Our discovery of HIF2 involvement opens new avenues for investigating the molecular basis of coronary vasculitis and may ultimately contribute to the development of targeted therapies taking advantage of HIF2-specific inhibitory treatments recently approved for clinical use.⁵⁵

ARTICLE INFORMATION

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Author Contributions

B. Escobar and Drs Menendez-Montes and Albendea-Gomez performed the experiments, analyzed the data, and helped write the manuscript. B. Palacios, S. Mendoza-Tamajon, R. Castro-Mecinas, S. Urra-Balduz, and Dr Díaz-Díaz performed the experiments. Drs Gómez and Sánchez-Cabo performed the RNA sequencing and functional analysis. Dr Ruiz-Cabello performed and analyzed magnetic resonance imaging and coronary resin cast filling using positron emission tomography-computed tomography. Dr Jiménez-Borreguero performed and analyzed the echocardiographic studies and coronary diameter analysis and provided essential cardiologic feedback for the project. Dr Cid provided critical discussions of the experiments and the manuscript. Dr Takahashi provided Kawasaki human tissue and critically discussed the results. Dr Martín-Puig defined the concept; planned, performed, and supervised the experiments; analyzed the data; wrote the manuscript; composed the figures together with Dr Albendea-Gomez; and obtained funding to support the study.

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REFERENCES

- Pouyssegur J, Dayan F, Mazure NM. Hypoxia signalling in cancer and approaches to enforce tumour regression. *Nature*. 2006;441:437–443. doi: 10.1038/nature04871
- Nakamura M, Sadoshima J. Mechanisms of physiological and pathological cardiac hypertrophy. *Nat Rev Cardiol*. 2018;15:387–407. doi: 10.1038/s41569-018-0007-y
- Tirpe AA, Gulei D, Ciortea SM, Crivii C, Berindan-Neagoe I. Hypoxia: overview on hypoxia-mediated mechanisms with a focus on the role of HIF genes. *Int J Mol Sci*. 2019;20:6140. doi: 10.3390/ijms20246140
- Huang X, Akgun EE, Mehmood K, Zhang H, Tang Z, Li Y. Mechanism of hypoxia-mediated smooth muscle cell proliferation leading to vascular remodeling. *Biomed Res Int*. 2022;2022:3959845. doi: 10.1155/2022/3959845
- Wan H, Liu D, Liu B, Sha M, Xia W, Liu C. Bioinformatics analysis of aging-related genes in thoracic aortic aneurysm and dissection. *Front Cardiovasc Med*. 2023;10:1089312. doi: 10.3389/fcvm.2023.1089312
- Chan XY, Volkova E, Eoh J, Black R, Fang L, Gorashi R, Song J, Wang J, Elliott MB, Barreto-Ortiz SF, et al. HIF2A gain-of-function mutation modulates the stiffness of smooth muscle cells and compromises vascular mechanics. *iScience*. 2021;24:102246. doi: 10.1016/j.isci.2021.102246
- Sluimer JC, Gasc JM, van Wanroij JL, Kisters N, Groeneweg M, Sollewijn Gelpke MD, Cleutjens JP, van den Akker LH, Corvol P, Wouters BG, et al. Hypoxia, hypoxia-inducible transcription factor, and macrophages in human atherosclerotic plaques are correlated with intraplaque angiogenesis. *J Am Coll Cardiol*. 2008;51:1258–1265. doi: 10.1016/j.jacc.2007.12.025
- Gnarra JR, Ward JM, Porter FD, Wagner JR, Devor DE, Grinberg A, Emmert-Buck MR, Westphal H, Klausner RD, Linehan WM. Defective placental vasculogenesis causes embryonic lethality in VHL-deficient mice. *Proc Natl Acad Sci U S A*. 1997;94:9102–9107. doi: 10.1073/pnas.94.17.9102
- Kaelin WG Jr. Von Hippel–Lindau disease: insights into oxygen sensing, protein degradation, and cancer. *J Clin Invest*. 2022;132:e162480. doi: 10.1172/JCI162480
- Tang N, Mack F, Haase VH, Simon MC, Johnson RS. pVHL function is essential for endothelial extracellular matrix deposition. *Mol Cell Biol*. 2006;26:2519–2530. doi: 10.1128/MCB.26.7.2519-2530.2006
- Menendez-Montes I, Escobar B, Palacios B, Gomez MJ, Izquierdo-Garcia JL, Flores L, Jimenez-Borreguero LJ, Aragonés J, Ruiz-Cabello J, Torres M, et al. Myocardial VHL-HIF signaling controls an embryonic metabolic switch essential for cardiac maturation. *Dev Cell*. 2016;39:724–739. doi: 10.1016/j.devcel.2016.11.012
- Martin-Puig S, Wang Z, Chien KR. Lives of a heart cell: tracing the origins of cardiac progenitors. *Cell Stem Cell*. 2008;2:320–331. doi: 10.1016/j.stem.2008.03.010
- Cai CL, Martin JC, Sun Y, Cui L, Wang L, Ouyang K, Yang L, Bu L, Liang X, Zhang X, et al. A myocardial lineage derives from Tbx18 epicardial cells. *Nature*. 2008;454:104–108. doi: 10.1038/nature06969
- Zhou B, Ma Q, Rajagopal S, Wu SM, Domian I, Rivera-Feliciano J, Jiang D, von Gise A, Ikeda S, Chien KR, et al. Epicardial progenitors contribute to the cardiomyocyte lineage in the developing heart. *Nature*. 2008;454:109–113. doi: 10.1038/nature07060
- Katz TC, Singh MK, Degenhardt K, Rivera-Feliciano J, Johnson RL, Epstein JA, Tabin CJ. Distinct compartments of the proepicardial organ give rise to coronary vascular endothelial cells. *Dev Cell*. 2012;22:639–650. doi: 10.1016/j.devcel.2012.01.012
- Lupu IE, Redpath AN, Smart N. Spatiotemporal analysis reveals overlap of key proepicardial markers in the developing murine heart. *Stem Cell Rep*. 2020;14:770–787. doi: 10.1016/j.stemcr.2020.04.002
- Duim SN, Kurakula K, Goumans MJ, Kruijthof BP. Cardiac endothelial cells express Wilms' tumor-1: Wt1 expression in the developing, adult and infarcted heart. *J Mol Cell Cardiol*. 2015;81:127–135. doi: 10.1016/j.jmcc.2015.02.007
- Duim SN, Smits AM, Kruijthof BP, Goumans MJ. The roadmap of WT1 protein expression in the human fetal heart. *J Mol Cell Cardiol*. 2016;90:139–145. doi: 10.1016/j.jmcc.2015.12.008
- Albendea-Gomez T, Mendoza-Tamajon S, Castro-Mecinas R, Escobar B, Ferreira Rocha S, Urrea-Balduz S, Nicolas-Avila JA, Oliver E, Villalba-Orero M, Martin-Puig S. Vascular HIF2 signaling prevents cardiomegaly, alveolar congestion, and capillary remodeling during chronic hypoxia. *Arterioscler Thromb Vasc Biol*. 2025;45:e78–e98. doi: 10.1161/ATVBAHA.124.321780
- Diaz Del Moral S, Barrena S, Hernandez-Torres F, Aranega A, Villaseca JM, Gomez Doblas JJ, Franco D, Jimenez-Navarro M, Munoz-Chapulí R, Carmona R. Deletion of the Wilms' tumor suppressor gene in the cardiac troponin-T lineage reveals novel functions of WT1 in heart development. *Front Cell Dev Biol*. 2021;9:683861. doi: 10.3389/fcell.2021.683861
- Jennette JC, Falk RJ, Andrasik K, Bacon PA, Churg J, Gross WL, Hagen EC, Hoffman GS, Hunder GG, Kallenberg CG. Nomenclature of systemic vasculitides: proposal of an international consensus conference. *Arthritis Rheum*. 1994;37:187–192. doi: 10.1002/art.1780370206
- Burns JC, Glode MP. Kawasaki syndrome. *Lancet*. 2004;364:533–544. doi: 10.1016/S0140-6736(04)16814-1
- Newburger JW, Takahashi M, Burns JC. Kawasaki disease. *J Am Coll Cardiol*. 2016;67:1738–1749. doi: 10.1016/j.jacc.2015.12.073
- Rowley AH, Baker SC, Shulman ST, Fox LM, Takahashi K, Garcia FL, Crawford SE, Chou P, Orenstein JM. Cytoplasmic inclusion bodies are detected by synthetic antibody in ciliated bronchial epithelium during acute Kawasaki disease. *J Infect Dis*. 2005;192:1757–1766. doi: 10.1086/497171
- Lehman TJ, Walker SM, Mahnovski V, McCurdy D. Coronary arteritis in mice following the systemic injection of group B *Lactobacillus casei* cell walls in aqueous suspension. *Arthritis Rheum*. 1985;28:652–659. doi: 10.1002/art.1780280609
- Takahashi K, Oharaseki T, Wakayama M, Yokouchi Y, Naoe S, Murata H. Histopathological features of murine systemic vasculitis caused by *Candida albicans* extract: an animal model of Kawasaki disease. *Inflamm Res*. 2004;53:72–77. doi: 10.1007/s00011-003-1225-1
- Noval Rivas M, Arditi M. Kawasaki disease: pathophysiology and insights from mouse models. *Nat Rev Rheumatol*. 2020;16:391–405. doi: 10.1038/s41584-020-0426-0
- Lee Y, Schulte DJ, Shimada K, Chen S, Crother TR, Chiba N, Fishbein MC, Lehman TJ, Arditi M. Interleukin-1 β is crucial for the induction of coronary artery inflammation in a mouse model of Kawasaki disease. *Circulation*. 2012;125:1542–1550. doi: 10.1161/CIRCULATIONAHA.111.072769
- Hui-Yuen JS, Duong TT, Yeung RS. TNF- α is necessary for induction of coronary artery inflammation and aneurysm formation in an animal model of Kawasaki disease. *J Immunol*. 2006;176:6294–6301. doi: 10.4049/jimmunol.176.10.6294
- Haase VH, Glickman JN, Socolovsky M, Jaenisch R. Vascular tumors in livers with targeted inactivation of the Von Hippel–Lindau tumor suppressor. *Proc Natl Acad Sci U S A*. 2001;98:1583–1588. doi: 10.1073/pnas.98.4.1583
- Gruber M, Hu CJ, Johnson RS, Brown EJ, Keith B, Simon MC. Acute postnatal ablation of Hif-2 α results in anemia. *Proc Natl Acad Sci U S A*. 2007;104:2301–2306. doi: 10.1073/pnas.0608382104

32. del Monte G, Casanova JC, Guadix JA, MacGrogan D, Burch JB, Perez-Pomares JM, de la Pompa JL. Differential Notch signaling in the epicardium is required for cardiac inflow development and coronary vessel morphogenesis. *Circ Res*. 2011;108:824–836. doi: 10.1161/CIRCRESAHA.110.229062
33. Li G, Zhang Y, Qian Y, Zhang H, Guo S, Sunagawa M, Hisamitsu T, Liu Y. Interleukin-17A promotes rheumatoid arthritis synovocytes migration and invasion under hypoxia by increasing MMP2 and MMP9 expression through NF- κ B/HIF-1 α pathway. *Mol Immunol*. 2013;53:227–236. doi: 10.1016/j.molimm.2012.08.018
34. Kietzmann T, Roth U, Jungermann K. Induction of the plasminogen activator inhibitor-1 gene expression by mild hypoxia via a hypoxia response element binding the hypoxia-inducible factor-1 in rat hepatocytes. *Blood*. 1999;94:4177–4185.
35. Chaurasia SN, Kushwaha G, Kulkarni PP, Mallick RL, Latheef NA, Mishra JK, Dash D. Platelet HIF-2 α promotes thrombogenicity through PAI-1 synthesis and extracellular vesicle release. *Haematologica*. 2019;104:2482–2492. doi: 10.3324/haematol.2019.217463
36. Kurozumi A, Nakano K, Yamagata K, Okada Y, Nakayama S, Tanaka Y. IL-6 and sIL-6R induces STAT3-dependent differentiation of human VSMCs into osteoblast-like cells through JMJD2B-mediated histone demethylation of RUNX2. *Bone*. 2019;124:53–61. doi: 10.1016/j.bone.2019.04.006
37. Bjerre M. Osteoprotegerin (OPG) as a biomarker for diabetic cardiovascular complications. *Springerplus*. 2013;2:658. doi: 10.1186/2193-1801-2-658
38. Simonini G, Masi L, Giani T, Piscitelli E, Cimaz R, Vierucci S, Brandi ML, Falcini F. Osteoprotegerin serum levels in Kawasaki disease: an additional potential marker in predicting children with coronary artery involvement. *J Rheumatol*. 2005;32:2233–2238.
39. Tirziu D, Chorianopoulos E, Moodie KL, Palac RT, Zhuang ZW, Tjwa M, Roncal C, Eriksson U, Fu Q, Elfenbein A, et al. Myocardial hypertrophy in the absence of external stimuli is induced by angiogenesis in mice. *J Clin Invest*. 2007;117:3188–3197. doi: 10.1172/JCI32024
40. Dai Z, Cheng J, Liu B, Yi D, Feng A, Wang T, An L, Gao C, Wang Y, Zhu MM, et al. Loss of endothelial hypoxia inducible factor-prolyl hydroxylase 2 induces cardiac hypertrophy and fibrosis. *J Am Heart Assoc*. 2021;10:e022077. doi: 10.1161/JAHA.121.022077
41. Kakkar R, Lee RT. Intramyocardial fibroblast myocyte communication. *Circ Res*. 2010;106:47–57. doi: 10.1161/CIRCRESAHA.109.207456
42. Porter KE, Turner NA. Cardiac fibroblasts: at the heart of myocardial remodeling. *Pharmacol Ther*. 2009;123:255–278. doi: 10.1016/j.pharmthera.2009.05.002
43. Nguyen SV, Claycomb WC. Hypoxia regulates the expression of the adrenomedullin and HIF-1 genes in cultured HL-1 cardiomyocytes. *Biochem Biophys Res Commun*. 1999;265:382–386. doi: 10.1006/bbrc.1999.1674
44. Nishida K, Watanabe K, Echigo S, Mayumi M, Nishikimi T. Increased plasma adrenomedullin levels in Kawasaki disease with coronary artery involvement. *Am J Med*. 2001;111:165–166. doi: 10.1016/s0002-9343(01)00781-1
45. Nomura I, Abe J, Noma S, Saito H, Gao B, Wheeler G, Leung DY. Adrenomedullin is highly expressed in blood monocytes associated with acute Kawasaki disease: a microarray gene expression study. *Pediatr Res*. 2005;57:49–55. doi: 10.1203/01.PDR.0000147745.52711.DD
46. Lau AC, Duong TT, Ito S, Yeung RS. Matrix metalloproteinase 9 activity leads to elastin breakdown in an animal model of Kawasaki disease. *Arthritis Rheum*. 2008;58:854–863. doi: 10.1002/art.23225
47. Chua PK, Melish ME, Yu Q, Yanagihara R, Yamamoto KS, Nerurkar VR. Elevated levels of matrix metalloproteinase 9 and tissue inhibitor of metalloproteinase 1 during the acute phase of Kawasaki disease. *Clin Vaccine Immunol*. 2003;10:308–314. doi: 10.1128/cdi.10.2.308-314.2003
48. Takeda K, Cowan A, Fong GH. Essential role for prolyl hydroxylase domain protein 2 in oxygen homeostasis of the adult vascular system. *Circulation*. 2007;116:774–781. doi: 10.1161/CIRCULATIONAHA.107.701516
49. Kabbaha S, Milano A, Aldeyab MA, Thorlund K. Infliximab as a second-line therapy for children with refractory Kawasaki disease: a systematic review and meta-analysis of randomized controlled trials. *Br J Clin Pharmacol*. 2023;89:49–60. doi: 10.1111/bcp.15547
50. Burns JC, Kone-Paut I, Kuijpers T, Shimizu C, Tremoulet A, Arditi M. Review: found in translation: international initiatives pursuing interleukin-1 blockade for treatment of acute Kawasaki disease. *Arthritis Rheumatol*. 2017;69:268–276. doi: 10.1002/art.39975
51. Yang J, Jain S, Capparelli EV, Best BM, Son MB, Baker A, Newburger JW, Franco A, Printz BF, He F, et al. Anakinra treatment in patients with acute Kawasaki disease with coronary artery aneurysms: a phase I/IIa trial. *J Pediatr*. 2022;243:173–180.e8. doi: 10.1016/j.jpeds.2021.12.035
52. Sato W, Yokouchi Y, Oharaseki T, Asakawa N, Takahashi K. The pathology of Kawasaki disease aortitis: a study of 37 cases. *Cardiovasc Pathol*. 2021;51:107303. doi: 10.1016/j.carpath.2020.107303
53. Kaelin WG Jr, Ratcliffe PJ. Oxygen sensing by metazoans: the central role of the HIF hydroxylase pathway. *Mol Cell*. 2008;30:393–402. doi: 10.1016/j.molcel.2008.04.009
54. McGettrick AF, O'Neill LAJ. The role of HIF in immunity and inflammation. *Cell Metab*. 2020;32:524–536. doi: 10.1016/j.cmet.2020.08.002
55. Choi WSW, Boland J, Lin J. Hypoxia-inducible factor-2 α as a novel target in renal cell carcinoma. *J Kidney Cancer VHL*. 2021;8:1–7. doi: 10.15586/jkcvhl.v8i1.170
56. Menendez-Montes I, Escobar B, Gomez MJ, Albendea-Gomez T, Palacios B, Bonzon-Kulichenko E, Izquierdo-Garcia JL, Alonso AV, Ferrarini A, Jimenez-Borreguero LJ, et al. Activation of amino acid metabolic program in cardiac HIF1- α -deficient mice. *iScience*. 2021;24:102124. doi: 10.1016/j.isci.2021.102124
57. Li B, Dewey CN. RSEM: accurate transcript quantification from RNA-seq data with or without a reference genome. *BMC Bioinf*. 2011;12:323. doi: 10.1186/1471-2105-12-323
58. Subramanian A, Tamayo P, Mootha VK, Mukherjee S, Ebert BL, Gillette MA, Paulovich A, Pomeroy SL, Golub TR, Lander ES, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc Natl Acad Sci U S A*. 2005;102:15545–15550. doi: 10.1073/pnas.0506580102
59. Villalba-Orero M, Garcia-Pavia P, Lara-Pezzi E. Non-invasive assessment of HFpEF in mouse models: current gaps and future directions. *BMC Med*. 2022;20:349. doi: 10.1186/s12916-022-02546-3
60. Cruz-Adalia A, Jimenez-Borreguero LJ, Ramirez-Huesca M, Chico-Calero I, Barreiro O, Lopez-Conesa E, Fresno M, Sanchez-Madrid F, Martin P. CD69 limits the severity of cardiomyopathy after autoimmune myocarditis. *Circulation*. 2010;122:1396–1404. doi: 10.1161/CIRCULATIONAHA.110.952820